Check for updates

Empirical Article

TOURISM ECONOMICS
The Business and Finance of Tourism and Recreation

Tourism Economics
2021, Vol. 0(0) 1–43
© The Author(s) 2021
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/13548166211049857
journals.sagepub.com/home/teu
SAGE

Does tourism activity reduce the size of the informal economy? Capturing long-term heterogeneous linkages around the world

Aldo Salinas
ESAI Business School, Universidad de Especialidades Espíritu Santo, Ecuador

Cristian Ortiz
ESAI Business School, Universidad de Especialidades Espíritu Santo, Ecuador

Pablo Ponce
Carrera de Economía and Centro de Investigaciones Sociales y Económicas, Universidad Nacional de Loja, Ecuador

Javier Changoluisa
ESAI Business School, Universidad de Especialidades Espíritu Santo, Ecuador

Abstract

This paper investigates the long-term and causal relationship between tourism activity and the informal economy in 76 countries from 1995 to 2015. We explore this relationship at the global level and by country group, using panel, co-integration techniques that indicate the existence of a long-run co-integration relationship between tourism and informal economy for the whole sample and at the level of country groups. Additionally, the paper analyzes the long-run coefficients of the model by using fully modified ordinary least square regressions (FMOLS). The results from FMOLS evidence a negative and significant impact of tourism on the informal economy at the global level and in high, upper-middle, and lower-middle income countries, but a positive link in low-income countries. However, the results reveal a heterogeneous long-run relationship within country groups. Also, the result of the Dumitrescu-Hurlin Granger causality test indicates bidirectional causality in the global sample, but the direction of causality varies by country group. The main policy implication derived from our findings suggests that in order to reduce the size of informal economy, policy-makers should foster tourism activities.

JEL Classification: : J01, L83, C23, O57, C00, C01

Corresponding author:
Cristian Ortiz, ESAI Business School, Universidad de Especialidades Espíritu Santo, Av. Samborondón 5, Samborondón 091650, Ecuador.
Email: cristian.ortiz@unl.edu.ec

Keywords
informal economy, tourism, panel data, co-integration, FMOLS

Introduction

The existence of informal economic activities is a global phenomenon, widespread especially in developing countries. Extensive cross-country research has shown that the prevalence of this type of economic activity has been greater in recent decades (Gerxhani, 2004; Schneider and Buehn, 2013; Schneider, 2015; Schneider, 2017; Williams and Schneider, 2016). Moreover, the level that the informal economy reaches varies considerably across countries at different levels of development (La Porta and Shleifer, 2014; Loayza, 2016). For example, estimates coming from Medina and Schneider (2018), show that the informal economy reaches an average size of 31.9% of the GDP in a sample of 158 countries over the period 1991–2015, and it is precisely the least developed countries that reach high percentages of informal economic activity.

Due to the significant share of the informal economy at the country level, its determinants have been the subject of extensive research in recent years. In the existing literature on the informal economy and its determinants, three approaches stand out. The first relates to the dual theory of development (Lewis, 1954). According to this perspective, the informal economy arises as a consequence of the failures of the economic system to generate sufficient employment in the formal sector, a product of the still deficient and backward productive structures of many economies (Banerjee and Duflo, 2011; La Porta and Shleifer, 2014). A second approach relates to the institutionalist view (De Soto, 1989; North 1990; Feige, 1990) that identifies excessive regulations and high tax burden as the key factors in explaining the high rates of informal activity in a country. Finally, the structuralist view considers the new logic of labor contracting and outsourcing as the most relevant determinants of the size of the informal economy. The restructure of world production processes would be linked to greater openness and deregulation of the world economy (Piore and Sabel, 1986; Castells and Portes, 1989).

Even though it is possible to note that tourism activities generate a series of productive dynamics that fit into the conceptual framework proposed by each of the three views on informality, none of these approaches is sufficiently explicit in recognizing a possible relationship between tourism and the informal economy (Lv, 2020). Indeed, in the framework of the new development strategies, suggested by the modernizing perspective and linked to the dual theory of development, tourism has been noticed as a key sector in the promotion of economic growth, and therefore as an effective instrument for poverty alleviation and modernization of economies (Alam and Paramati, 2016; Croes, 2014; Hall, 2007; Truong, 2013).

In addition, several findings based on specific cities and countries show that the presence of large flows of tourists, especially in urban centers, generate entrepreneurial opportunities for street vendors, who work outside regulations on the use of public space and avoid paying taxes (Boonjubun, 2017; Truong, 2018). Likewise, the digital revolution and the emergence of sharing economy platforms have generated entrepreneurship opportunities in the tourism sector. For example, in the platform Airbnb many economic transactions are not subject to regulations of public authorities and there are serious indications of tax evasion (Guttentag, 2015), especially in less developed countries that have weak tax regulations. Also, the significant growth of tourism worldwide in recent decades has developed simultaneously with the emergence of tourism global production networks (Adiyia et al., 2015; Christian, 2016; Romero et al., 2020; Song et al., 2013). Several papers have found linkages between large tourism firms and small local firms through

outsourcing, subcontracting and other types of informal relationships (Ashley, 2006; Kirsten and Rogerson, 2002; Mshenga and Richardson, 2013).

The previous discussion suggests the existence of a heterogeneous link between tourism and informal economy. However, previous research that explored this relationship at the empirical level shows evidence limited to a single country. Thus, it is relevant to contribute to the literature by exploring the relationship between tourism and the informal economy by a cross-country and cross-temporal study. To the best of our knowledge, it is among the first studies to explore this relationship at the global level and by country group, and especially considering a long-run perspective. Exceptions are the contributions recently made by Lv (2020) and Xu and Lv (2021), who, however, study this relationship based on a short time period and a different econometric approach. Our approach of studying a possible long-run relationship between these two variables is more appropriate, as informal activity rates at the country level, seem to be persistent in the short-run (Loayza, 2016). Also, a dynamic rather than static framework is more suitable. Therefore, the limitation of few existing previous studies suggests that further exploration is needed. Additionally, the three views of informality do not allow us to anticipate what could be the sign of a possible relationship between tourism activity and informal economy.

From this perspective, the relevant contribution of this paper is to determine the heterogeneous long-run and causal relationship between a measure of tourism activity and the informal economy in 76 countries over the period 1995 to 2015; both at the global and country group level. To capture the heterogeneity of growth, we use the Atlas classification proposed by the World Bank, such that we classify the sample countries into four groups: high-income countries (HICs); Upper-middle income countries (UMICs), lower-middle income countries (LMICs), and low-income countries (LICs). We use the informal economy data presented by Medina and Schneider (2018), who estimate the size of the informal economy using a structural equation model. This measure includes all economic activities hidden from official authorities for monetary, regulatory, and institutional reasons (Medina and Schneider, 2018). We chose to use this measure given the robustness of their method, and geographical and long-period coverage. In regards to our measure of tourism activity, we consider a tourism activity index (TI) compounded of three variables that allows us to capture the multidimensional nature of tourism. In addition, we use some control variables such as GDP per capita, tax burden and a measure of the rule of law.

Therefore, we employ panel co-integration techniques to test the long-run and causal relationship between tourism and informal economy. After testing the presence of cross-sectional dependence and slope heterogeneity, we use panel co-integration test proposed by Westerlund (2007). The results show evidence of a long-run co-integrating relationship between tourism and informal economy. Next, we used a fully modified ordinary least squares (FMOLS) model to investigate long-run impacts of the explanatory variables. The individual FMOLS coefficients indicate that tourism has a negative but significant impact on the informal economy at the global level and in HICs, UMICs, and LMICs, whereas that tourism increases the informal economy in low-income countries. Finally, the Dumitrescu and Hurlin (2012) heterogeneous panel causality test shows evidence of significant bidirectional causality relationship between tourism and the informal economy for the global, UMICs, and LMICs panels. Also, we note the presence of unidirectional causality running from tourism to the informal economy for LICs and from the informal economy to tourism for HICs.

This paper is organized in four sections. Literature review contains a comprehensive review of previous literature. Data and statistical strategy is divided into two sections: Data contains the description of the data and variables; and Statistical strategy details the statistical strategy used.

Discussion of results contains the discussion of our findings. Finally, Conclusions and policy implications contains the main conclusions and policy implications of our findings.

Literature review

Research on the informal economy has been broad and diverse. In the literature, there are three views that have prevailed in the debate on the informal economy (De Paula and Scheinkman, 2011). The first relates to the dual theory of economic development (Lewis, 1954). From this view, informal activities are temporary and undertaken out of necessity, due to the lack of employment in the formal sector (Gindling and Newhouse, 2014; Margolis, 2014). Therefore, these activities are highly inefficient, and make use of low human capital and primitive technologies (La Porta and Shleifer, 2008, 2014). The main conclusion derived from this view is that informality is a sign of underdevelopment. Then, the size of the informal economy is inversely related to the level of economic development (Elgin and Oztunali, 2014).

In the same vein, from the dual view of informality, there is a clear link between poverty and informality (Nikopour and Shah Habibullah, 2010). For example, several papers find that tourism firms operating in the informal sector are very different as compared to formal firms in certain characteristics such as human capital, management skills and wages paid (Liu and Wall, 2006). Likewise, previous studies find that much of these tourism activities are merely for survival purposes, rarely grow and often supplement the income of poor households (Damayanti et al., 2017, 2018; Lashley and Rowson, 2010; Mshenga et al., 2010). For example, during economic downturns, the informal economy is a key option for some street vendors, unofficial guides, individual transport providers, handicraft producers, artisans, boat keepers, bicycle rentals, food stall owners, souvenir vendors, and so on, who are located near tourist attractions (Çenesiz and Çakmak, 2020; Onodugo et al., 2016).

The dual view of the informal economy is closely related to modernization theories, which suggest that economic development occurs primarily through Big Pushes (Murphy et al., 1989; Rostow, 1990). Within this new framework for development, tourism has been noticed as a key sector in promoting economic growth, and thus as an instrument of poverty alleviation (Hall, 2007; Truong, 2013, 2014) and modernization of the economy (Alam and Paramati, 2016). Several papers have found a positive relationship between tourism and economic growth (Eugenio-Martin et al., 2004; Fayissa et al., 2009; Sequeira and Maçãs Nunes, 2008; Pulido-Fernández and Cárdenas-García, 2021). That is, applying the theoretical framework of dual development theory, we would expect large flows of tourism activity to generate opportunities for the creation of formal firms, operated by entrepreneurs with better human capital and management skills. In turn, it will be these new firms that will absorb labor from the informal sector (Banerjee and Duflo, 2011; La Porta and Shleifer, 2014).

The second view is related to the conceptual framework proposed from institutional economics (North 1990). From this approach, the institutions or rules of the game are what determine the evolution and composition of the formal and informal sectors (De Soto, 1989, 2000; Feige, 1990; Perry and Maloney, 2007). From an institutionalist view, the informal economy is related to a set of firms, workers and activities that operate outside formal institutional boundaries, but within informal institutional boundaries (Webb et al., 2014; Welter et al., 2015). Formal boundaries are related to the regulatory and legal framework (Loayza, 2016). For example, empirical evidence suggests that a high regulatory and tax burden incentivizes participation in the informal sector (Enste, 2010; Schneider et al., 2010). On the other hand, informal boundaries correspond to the

"socially shared rules, usually unwritten, that are created, communicated, and enforced outside of officially sanctioned channels" (Helmke and Levitsky, 2004: p. 727)

In recent years, some scholars have been suggesting that the greater the incoherence between formal and informal rules the more entrepreneurs will operate in the informal sector (Damayanti et al., 2017; De Castro et al., 2014; Webb et al., 2009, 2014; Welter and Smallbone, 2011; Williams et al., 2016; Williams and Vorley, 2014). A good example of this incongruity in the tourism sector is the emergence of sharing economy platforms, such as Airbnb (Alrawadieh and Alrawadieh, 2018; Williams and Horodnic, 2017). Much of the economic transactions generated through the Airbnb platform are not subject to regulations from public authorities, and there are indications of tax evasion (Guttentag, 2015). However, at the level of informal rules, there is a legitimization of the activity among the population, which is manifested by the growing popularity of the platform. Likewise, the use of urban space by street vendors located near tourist attractions is another example of incongruities between formal and informal rules (Boonjubun, 2017; Chavarria and Phakdee-auksorn, 2017; Truong, 2018; Yeo and Heng, 2014). Although local governments regulate the use of public space such as squares and streets near tourist attractions, these spaces are occupied by informal economy actors, who may claim a right over particular locations, for appealing to a set of norms, values, and beliefs that legitimize their actions (Çakmak et al., 2018; Damayanti et al., 2018; Onodugo et al., 2016; Pécot et al., 2018).

A third perspective relates to the expansion of the informal economy to the process of global production restructuring that began during the 1970s, as a consequence of further opening and deregulation of the world economy (Castells and Portes, 1989). The new production paradigm implied moving from a vertically integrated mass production system to a flexible production system (Piore and Sabel, 1986), where parts of the production process can be subcontracted or outsourced to other firms, located in different parts of the world, and linked through global production networks (Coe and Yeung, 2015; Gereffi, 1999). In the tourism sector, there are tourism global production networks, that includes distribution (tour operators), transportation (air, ground), accommodation, and broad tourist-related activities (Clancy, 2011; Christian, 2016; Murphy, 2019; Romero et al., 2020). Under this new logic of flexible and deregulated production, the informal economy becomes one more linked in these production networks. In other words, there is a symbiotic relationship between the formal and informal economy, where some production processes are outsourced to small informal enterprises (Jones et al., 2006). In addition, many workers and small informal enterprises develop situations of dependence or subordination compared to large formal companies (Chen, 2006; Guha-Khasnobis et al., 2006).

Regarding tourism, important links between formal and informal activity have been recognized (Çakmak et al., 2018; Henderson and Smith, 2009). For example, several multinational tourism companies hire salaried workers under informal employment relationships (Bianchi and De Man, 2021; Reid, 2003) and, many tourism companies hire migrant labor under poor working condition (Baum and Hai, 2019; Duncan et al., 2013; Janta et al., 2012). In the same way, productive synergies have been found among large-scale resorts and informal tourism commerce (Henderson and Smith, 2009). On the other hand, several works warn of the weakness of many developing countries to enforce regulations to multinational companies. In fact, through illicit transaction, locally-based elites seek to incorporate their countries into global networks specially related to the ecotourism industry (Duffy, 2000; Mbaiwa and Hambira, 2020). In addition, other work has found linkages between large tourism firms and small local firms through outsourcing, subcontracting and other types of informal arrangements (Anderson and Juma, 2011; Ashley, 2006; Kirsten and Rogerson, 2002; Mshenga and Richardson, 2013; Romero et al., 2020). Likewise, informal employment is common in restaurants and hotels (Chen, 2006). This evidence suggests, then, that increased

tourism development generates opportunities for entrepreneurship in the informal sector through linkages with the formal tourism sector.

The previous discussion highlights some issues. First, the dual view of informality, related to Big Pushes theories, suggests a negative relationship between tourism development and the informal economy. The underlying argument is that a greater influx of tourists generates larger entrepreneurship and investment opportunities in the formal sector, which in turn are recognized by entrepreneurs with better human capital and management skills. In turn, these formal firms will absorb labor from the informal sector (La Porta and Shleifer, 2008). Second, the institutionalist perspective suggests that the sign of the relationship will depend on the quality of business regulation in each country. Better regulation will encourage tourism entrepreneurs to prefer to operate in the formal rather than the informal sector. Finally, the structuralist perspective suggests a positive relationship between tourism and informality. The presence of tourism global production networks can lead to subcontracting relationships with local informal micro-enterprises, or hiring workers at low wages with few benefits, many of them part-time workers, or temporary workers (Clancy, 2011; Reid, 2003). In summary, the relationship between tourism and informality becomes an empirical problem that we disentangle in this paper.

Data and statistical strategy

Data

In this research, we constructed a panel data covering the time period 1995–2015 that includes 1596 observations in 76 countries. The selection of these countries is based on data availability. The dependent variable is the size of the informal economy. We utilize the data from Medina and Schneider (2018), who estimate the size of the informal economy using indirect estimation methods. Specifically, they use a special type of structural equation modeling to make estimates for about 158 countries. The measure includes all economic activities that are hidden from official authorities for monetary, regulatory, and institutional reasons (Medina and Schneider, 2018). The database has been widely used in the literature of informality (see Baklouti and Boujelbene, 2020; Colombo et al., 2019; Huynh and Nguyen, 2020; Gutiérrez-Romero, 2021; Lv, 2020). Also, we use Medina and Schneider (2018) estimates as they offer a broad geographical and temporal coverage.

Next, as a measure of tourism activity, we follow Snieška and Bruneckiene (2009) and Ponce et al. (2019), [1] and build a tourism activity index (TI), which is a relativized measure estimated from three variables that are the most commonly used measures of tourism demand (see Peng et al., 2014, 2015): tourism expenditures (TE, measured in current US dollars), the total number of tourist arrivals (TA) and tourism receipts (TR, measured in current US dollars). According to Ponce et al. (2019) constructing a relativized index allows for a more efficient aggregate estimation of different variables in a single measure, thus capturing the multidimensional essence of the phenomenon studied. The calculation formula is given by

$$\mathrm{TI}_i = 1/f \sum_{i}^{n} \mathrm{Sub\ index}_i \tag{1}$$

Where $i = 1..., n$; $f = 1, ..., 3$; therefore, TI_i is the tourism performance of country i, $\mathrm{Sub\ index}_i$ is the factor index of f in country i. That is, TI is constructed with the average values of each sub index. The sub index of each variable is calculated in such a way that the maximum possible value is 10 and the lowest possible value is 1, and the interval separating the intermediate countries is equidistant in relation to the extremes depending on the level of tourism activity, thus relativizing the value

obtained. Then, we subtract the variable with the minimum value of all the countries from the variable j of country i and fraction by the difference between the variables with the maximum and minimum value as follows

$$\text{Sub index}_i = \left[9 \times \frac{\text{Value}_{ji} - \text{Minimum value}_i}{\text{Maximum value}_i - \text{Minimum value}_i}\right] + 1 \quad (2)$$

In addition, as a robustness check, we use a different approach to build the tourism activity index (TAI). This alternative approach corresponds to the principal component analysis that allow us to build a weighted tourism index of all the three tourism indicators (see Lv, 2020; Zaman et al., 2016). This method ranked the importance of causal factors and reduced the dimensionality of the factors by excluding their correlations (Wang et al., 2016). The results of the principal component analysis for tourism index are reported in Table B1 (Appendix B). On the basis of these results, we extract the principal component factor loading (PC1), which capture 90.27% of the information. Moreover, this component is the only one with an eigenvalue greater than one, therefore, we used PC1 factor loading to build the tourism activity index.

Finally, we control for the usual determinants that have been noted in the literature. Specifically, we consider three control variables: GDP per capita, rule of law, and taxation. Several studies have found a negative relationship between the informal economy and economic development (Autio and Fu, 2015; La Porta and Shleifer, 2008), for this reason, we have used GDP per capita as a control variable. Likewise, we also control for a measure of rule of law, given that several papers have found that those countries with a stronger rule of law, present a smaller size of their informal economy (Dau and Cuervo-Cazurra, 2014; Salinas et al., 2019). Finally, following the arguments of Johnson et al. (1998) and Friedman et al. (2000), we also control for the level of taxation in each country, since a high tax burden is likely to foster the informal economy (Webb et al., 2013). Table 1 details a brief description, measure, and source of the data.

Table 1. Description of variables and data sources.

Variable	Symbol	Description	Measure	Data source
Informal economy	IEC	It is the size of the informal economy as a percentage of GDP. Normalized between 0 and 1	Percent	Medina and Schneider (2018)
Tourism index	TI	It is a relativized measure estimated from three variables: Tourism expenditures, the total number of tourist arrivals, and tourism receipts Normalized between 1 and 9 1 represents the lowest level of tourism and 10 the highest level of tourism	Index	World Bank
Log GDP	LogGDP	Log GDP per capita (in constant 2010)	US dollar	World Bank
Taxation	TAX	It is a measure of tax revenue within each country Normalized between 0 and 1	Percent of GDP	World Bank
Rule of law	RL	It is a measure of the strength and fairness of the legal system, and an assessment of popular compliance with the law Normalized between 0 and 1. A higher value indicates stronger rule of law	Index	International country risk guide (ICRG)

Since the literature has identified that the effects of a variable can change considerably between different groups of countries according to their income or development level (Chrid et al., 2020; Ortiz et al., 2019), the 76 countries are classified based on the income level of the countries, and following the World Bank Atlas method of year 2020, which classifies countries into four groups: High, Upper-Middle, Lower-Middle, and Low-Income. This classification has been widely used in previous research (Alvarado et al., 2020; Chrid et al., 2020; Wang et al., 2018).

Table 2 shows the descriptive statistics and the correlation matrix of the variables used in the statistical estimates. Panel A of Table 2 shows that the variables form a balanced panel with 1596 observations, within a period of 21 years (T = 21) and the total number of countries in our sample is 76 (n = 76). The informal economy variable shows greater variability over time between countries than within countries, the standard deviation (SD) between countries is 0.126 and within countries is 0.035. Similarly, the tourism activity index shows greater variability over time between countries (SD = 1.529) than within countries (SD = 0.479), indicating that most of the variance comes from variations between countries. Additionally, panel B of Table 2 shows the correlation matrix between the variables, where we can observe that all the independent variables are negatively and statistically related with our variable that measures informal economy IEC.

Figure 1 shows the evolution of the variables by income level. It is possible to observe that the informal economy variable has experienced a slight decrease in all country groups in recent years. As expected, high-income countries have lower levels of informal economy, while low-income countries have higher levels of informal economy. On the other hand, the tourism activity seems to

Table 2. Descriptive statistics and correlation matrix of the variables.

	IEC	TI	logGDP	TAX	RL
Panel A: Descriptive statistics					
Mean	0.268	5.974	9.112	23.491	4.362
Std. Dev. (Overall)	0.130	1.594	1.380	2.011	1.229
Std. Dev. (Between)	0.126	1.529	1.371	1.998	1.144
Std. Dev. (Within)	0.035	0.479	0.202	0.315	0.467
Min	0.061	1.842	5.212	18.690	2.634
Max	0.719	9.972	11.625	28.259	6.319
N	1596	1596	1596	1596	1596
n	76	76	76	76	76
T	21	21	21	21	21
Panel B: Correlation matrix					
SEC	1				
	–				
TI	−0.548***	1			
	(0.010)	–			
logGDP	−0.713***	0.536***	1		
	(0.009)	(0.000)	–		
TAX	−0.518***	0.511***	0.528***	1	
	(0.000)	(0.000)	(0.000)	–	
RL	−0.6331***	0.2908***	0.5008***	0.3604***	1
	(0.000)	(0.000)	(0.000)	(0.000)	–

The asterisks mean: *p < 0.05, **p < 0.01, ***p < 0.001.

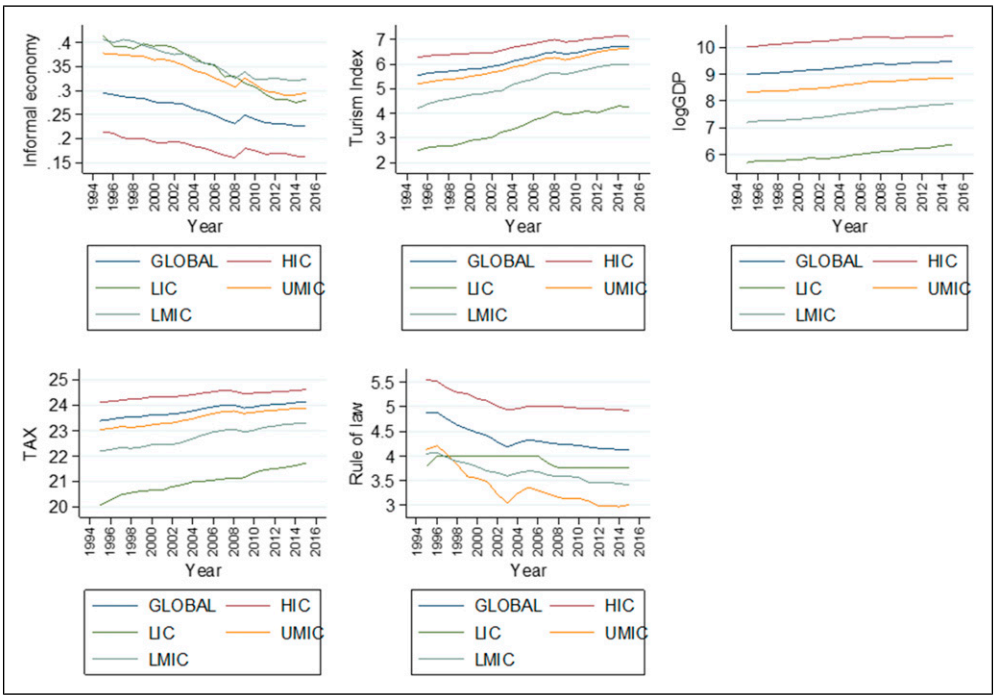


Figure 1. Evolution of variables by income level.

have experienced an increase at all income levels. This positive trend can also be observed in the GDP per capita variable and in the tax burden, which has increased over the period analyzed. Lastly, the rule of law variable shows a negative trend at the global level and by country group.

Figure 2 shows a map of quantiles of a country's level of informal economy and its respective tourism activity index. In this map, we show that the informal economy levels tend to be higher in relatively lower income countries. In general, this picture shows that higher income countries have lower levels of informal economy and high levels of tourism activity. To further clarify this pattern, in Figure A1 (Appendix A), we present a correlation graph between the informal economy and the tourism activity index for the whole sample. The scatter plot confirms a negative relationship between the variables.

Statistical strategy

In order to analyze the effect of tourism activity on the informal economy, we used various econometric modeling techniques. As a preliminary step, we use Generalized Least Square (GLS) panel estimations, before looking into co-integration relationship. The following baseline model is used to estimate the tourism-informality nexus

$$\mathrm{IEC}_{i,t} = \beta_0 + \beta_1 TI_{i,t} + \beta_2 \mathrm{logGDP}_{i,t} + \beta_3 \mathrm{TAX}_{i,t} + \beta_4 \mathrm{RL}_{i,t} + \ \varepsilon_{i,t} + u_t \tag{3}$$

Where $\mathrm{IEC}_{i,t}$ represents the size of the informal economy of country i in period t (in percent of GDP). For its part, β_0 represents the constant. Meanwhile, $\beta_1, \beta_2, \beta_3$, and β_4 measure the effect of

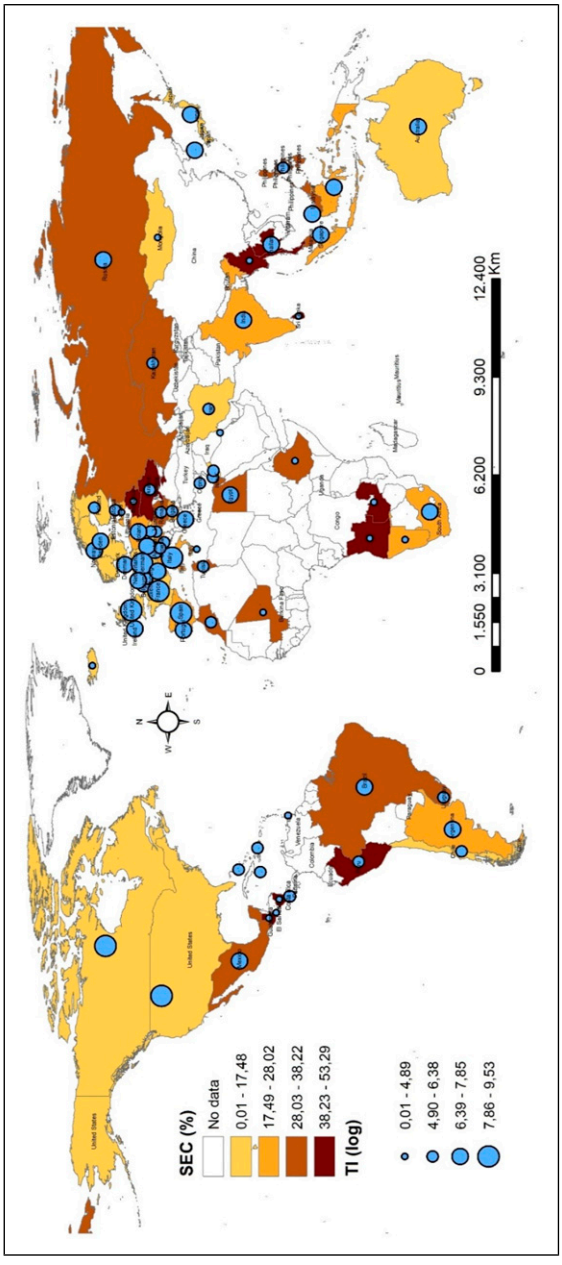


Figure 2. Informal economy and tourism index in the world.

tourism activity, GDP per capita, tax burden, and rule of law on the informal economy, respectively; $\varepsilon_{i,t}$ represents the error term and u_t measures the effect of an unobservable and time invariant term.

In order to further examine the long-run and causal relationships between tourism and the informal economy, we utilized multivariate panel data co-integration techniques. As a first step, we test the cross-sectional dependence across the countries. After, we analyze the homogeneity of the slope coefficients to take into account the characteristics of country-specific. Next, given the presence of cross-section dependence, we employed unit root tests of Pesaran (2007) to find the integration order of the series. In the fourth step, to verify the existence of a long-run relationship, a second-generation panel co-integration tests developed by Westerlund (2007) is employed. After the presence of co-integrating relationship is confirmed, the long-run coefficients can be estimated through a fully modified ordinary least squares (FMOLS) estimator (Pedroni, 2001). Lastly, we estimate the existence and direction of causality using the panel Granger non-causality technique. To do this, we use the statistical test proposed by Dumitrescu and Hurlin (2012).

Cross-sectional dependence tests. In this investigation, we use the cross-sectional dependence tests proposed by Pesaran (2004) to verify possible "spillover" effects, since changes over time, which happen in a variable in a given country, can generate an effect on the values of the same variable, but in the rest of the countries. The test verifies the null hypothesis that there is no cross-sectional dependence. This test is defined by

$$\mathrm{CD} = \sqrt{\frac{2T}{N(N-1)}} \left(\sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \widehat{\rho}_{ij} \right) \tag{4}$$

Also, we use the weak cross-sectional dependence test proposed by Pesaran (2015). This test measures the strength of cross-sectional dependence of the errors. The test is then defined by

$$\mathrm{CD} = \left[\frac{\mathrm{TN}(N-1)}{2} \right]^{\frac{1}{2}} \tilde{\rho} N \tag{5}$$

Where

$$\tilde{\rho} N = \frac{2}{N(N-1)} \sum_{i=1}^{N} \sum_{j=i+1}^{N} \widehat{\rho}_{ij} \tag{6}$$

Slope homogeneity tests. After checking cross-sectional dependence, we use the slope homogeneity test for panel data proposed by Pesaran and Yamagata (2008). Specifically, this test analyzes whether there is sufficient statistical evidence to affirm that the slopes (regression parameters) are heterogeneous, that is, there is heterogeneity among the coefficients of each country in our sample. The test statistic is given by

$$\tilde{\Delta} = \frac{1}{\sqrt{N}} \left(\frac{\sum_{i=1}^{n} \widetilde{d}_i - k_2}{\sqrt{2k_2}} \right) \tag{7}$$

Where the test statistic, under H0 in equation (7), is asymptotically $\tilde{\Delta} \sim N(0, 1)$. In equation (7), $\widetilde{d}_i$ is defined as the weighted difference between the cross-sectional unit-specific estimate and the pooled estimate. In the case of normally distributed errors, the adjusted mean-variance bias $\tilde{\Delta}$ can be expressed as follows

$$\tilde{\Delta}_{\text{adj}} = \sqrt{N}\left(\frac{N^{-1}\sum_{i=1}^{n}\widetilde{d}_i - k_2}{\sqrt{\text{Var}\left(\widetilde{z}_i, T_i\right)}}\right) \tag{8}$$

Where $\text{Var}(\widetilde{z}_i, T_i) = \frac{2k_2(T_i - K - 1)}{T_i - K + 1}$. Therefore, we can test whether the slopes are homogeneous across the individual units.

Panel unit root tests. If the presence of cross-sectional dependence is identified, we employ the unit root tests called Cross-sectional Augmented IPS (CIPS) and Cross-sectional Augmented Dickey-Fuller (CADF) proposed by Pesaran (2007), which allow us to control for cross-sectional dependence. This process helps us confirm the degree of stationary between the selected series across the various cross-sections. Thus, when there is dependence in the cross-sections, equation (9) formalizes the unit root test for CADF

$$\Delta\text{IEC}_{it} = \alpha_i + \phi_i\text{IEC}_{i\,t-1} + \beta_i\overline{\text{IEC}}_{t-1} + \sum_{j=0}^{k}\theta_{ij}\Delta\overline{\text{IEC}}_{it-j} + \sum_{j=0}^{k}\gamma_{ij}\text{IEC}_{it-1} + \varepsilon_{it} \tag{9}$$

Where, $\overline{\text{IEC}}_{t-1}$ is the cross-sectional mean at time t, k is the optimal lag length determined by information criteria of Akaike (1974), ε_{it} is the error term. Also, Pesaran (2007) proposed Cross-sectional Augmented IPS (CIPS), which can be calculated by the following equation

$$\text{CIPS} = \frac{1}{N}\sum_{i=1}^{N}\text{CADF}_i \tag{10}$$

Panel co-integration tests. In order to verify the existence of panel co-integration, we use the second-generation Westerlund (2007) panel co-integration test, which provides a methodological framework to test for the presence of long-run relationships while accounting for cross-sectional dependence in panel data. In this test, the null hypothesis is no co-integration, as opposed to the alternative hypothesis of co-integration, and it helps us to show whether a series has vectors (variables) that move simultaneously in the long-run. This test is based on structural rather than residual dynamics and shows good small sample properties. The model specification is given as

$$\Delta\text{IEC}_{it} = \delta_t^{'}d_t + \alpha_i\left(\text{IEC}_{it-1} - \varphi_i^{'}x_{it-1}\right) + \sum_{j=1}^{pi}\alpha_{ij}\Delta IEC_{it-j} + \sum_{j=-qi}^{pi}\gamma_{ij}\Delta x_{it-1} + \varepsilon_{it} \tag{11}$$

where d_t denotes the deterministic components, x_{it} is the vector of explanatory variables, ρ_i denotes the lag order, and α_{ij} is the error-correction parameter.

Panel co-integration regressions. After the presence of co-integrating relationship is confirmed, the long-run coefficients can be estimated through a FMOLS estimator (Pedroni, 2001). The FMOLS

method produces estimators that are asymptotically unbiased and standard normal distributions that are free of nuisance parameters (Wu and Xie, 2020). Also, FMOLS uses a semiparametric correction for endogeneity and residual autocorrelation, and the FMOLS estimator also allows for a high degree of panel heterogeneity (Liddle, 2012). Indeed, one of the main advantages of this approach is that it is possible to observe a different coefficient for each country analyzed in our sample. FMOLS estimator is constructed as follows

$$\widehat{\beta}^{*}_{\mathrm{GFM}} = N^{-1}\sum_{I=1}^{N}\left(\sum_{I=1}^{T}\left(X_{it}-\overline{X}_{i}\right)^{2}\right)^{-1}\times\left(\sum_{I=1}^{T}\left(X_{it}-\overline{X}_{i}\right)\mathrm{IEC}^{*}_{it}-T\widehat{\gamma}_{i}\right) \tag{12}$$

Where

$$\mathrm{IEC}^{*}_{it} = \left(\mathrm{IEC}_{it}-\overline{\mathrm{IEC}}_{i}\right)-\frac{\widehat{\Omega}_{21i}}{\widehat{\Omega}_{22i}}\,\Delta X_{it}$$

$$\widehat{\gamma}_{i} = \widehat{T}_{21i}+\widehat{\Omega}^{o}_{21i}-\frac{\widehat{\Omega}_{21i}}{\widehat{\Omega}_{22i}}\left(\widehat{T}_{22i}+\widehat{\Omega}^{o}_{22i}\right)$$

Panel causality test. Lastly, we analyze the causal relationships between the variables and estimate the existence and direction of causality for the Granger-type panel data. To do this, we use the statistical test proposed by Dumitrescu and Hurlin (2012). Compared to a panel vector error-correction model (VECM), this test allows both the heterogeneity and cross-sectional dependence. Also, the causality tests of Dumitrescu and Hurlin (2012) does not require N to be of a size larger than T. The causality tests of Dumitrescu and Hurlin (2012) are employed between pairs of variables, such that it tests whether allow unidirectional or bidirectional causality between all our four variables considered in this research. Equation (13) formalizes the Dumitrescu-Hurlin panel causality test as follows

$$\Delta\mathrm{IEC}_{it} = \alpha_{i}+\sum_{k=1}^{K}\gamma_{i}^{(k)}\Delta\mathrm{IEC}_{it-k}+\sum_{k=1}^{K}\beta_{i}^{(k)}\Delta x_{it-k}+\varepsilon_{it} \tag{13}$$

Where, α_i represents the constant vector; $\gamma_i^{(k)}$ is the lag parameter; $\beta_i^{(k)}$ is the slope of the coefficient, and k denotes the lag length. The null hypothesis of this test is that there is no causal relationship in the panel, whereas the alternative hypothesis is that there is a causal relationship in at least one cross-section unit. To test the null hypothesis, Dumitrescu and Hurlin calculated the Wald test statistic for all panel by taking the average of the individual Wald statistics. The test statistic is

$$W^{Hnc}_{N,T} = \frac{1}{N}\sum_{i=1}^{N}W_{i,T} \tag{14}$$

Discussion of results

First, in Table 3 we present the results of the models at the global level and by country groups using a Generalized Least Squares (GLS) estimator to correct the autocorrelation and heteroscedasticity problem, taking into account that the test proposed by Wooldridge (1991) and the modified Wald

test detected autocorrelation and heteroscedasticity, respectively, in all models. Looking at the overall data sample we find a negative impact of tourism on the informal economy. These findings hold for the case of the HICs, UMICs and LMICs. However, a positive impact of tourism activity on the size of the informal economy is evident in the case of low-income countries. Lv (2020) also evidenced the negative relationship between tourism activity and the informal economy at the global level and for the case of HICs, UMICs, and LMICs. Though Lv (2020) does not make the estimates for low-income countries.

In panel A of Table 4, the results of cross-sectional dependence test of Pesaran (2004) are shown. The results suggest that the null hypothesis of cross-sectional independence of the sections should be rejected for all variables in the model at 1% significance. Therefore, cross-sectional dependence

Table 3. Baseline regressions using GLS estimator.

	GLOBAL	HICs	UMICs	LMICs	LICs
TI	−0.0037*	−0.0025	−0.0141*	−0.0145***	0.0932***
	(0.0018)	(0.0024)	(0.0061)	(0.0041)	(0.0167)
logGDP	−0.1295***	−0.1389***	−0.0454*	−0.1243***	−0.2657***
	(0.0050)	(0.0062)	(0.0215)	(0.0098)	(0.0342)
TAX	−0.0504**	−0.0192	−0.0166	−0.1203*	−0.7883***
	(0.0166)	(0.0195)	(0.0521)	(0.0502)	(0.0860)
RL	−0.0019*	0.0017	0.0040	−0.0045*	−0.0259
	(0.0009)	(0.0010)	(0.0040)	(0.0020)	(0.0168)
Constant	1.4681***	1.6130***	0.8363***	1.4943***	1.7066***
	(0.0395)	(0.0632)	(0.1580)	(0.0778)	(0.2016)
Observations	1596	861	273	420	42
N	76	41	13	20	2
X^2	28540.48	11720.52	1996.17	13838.5005	2827.16
Year fix effects	Yes	Yes	Yes	Yes	Yes
Country fix effects	Yes	Yes	Yes	Yes	Yes

The dependent variable is informal economy. The asterisks in the parameters mean: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

Table 4. Cross-sectional dependence test and slope homogeneity test.

Panel A: CD test. Cross-section dependency tests of Pesaran (2004)					
Variables	IEC	TI	logGDP	TAX	RL
Test statistic	25.78***	6.13***	9.24***	23.76***	4.34***
p-value	0.000	0.0000	0.0000	0.0000	0.0000
Panel B: CD test. Test of Pesaran (2015) for weak cross-sectional dependence					
Variables	IEC	TI	logGDP	TAX	RL
Test Statistic	72.04***	32.22***	33.11***	44.21***	55.87***
p-value	0.000	0.0000	0.0000	0.0000	0.0000
Panel C: Slope homogeneity tests	$\tilde{\Delta}$	$\tilde{\Delta}_{adj}$			
Test Statistic	25.91***	30.38***			
p-value	0.000	0.000			

The asterisks in the parameters mean: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

exists in our research. That is, a shock in any of the countries can also affects other countries. Also, panel B of Table 4 presents the test of Pesaran (2015) for weak cross-sectional dependence. This test suggests rejecting the null hypothesis that the errors are weakly cross-sectional dependent. Therefore, the cross-sectional dependence is strong. Regarding to the slope heterogeneity problem, panel C of Table 4 shows the results of the homogeneity of slope tests proposed by Pesaran and Yamagata (2008). The p-value results of $\tilde{\Delta}$ and $\tilde{\Delta}_{\text{adj}}$ allow us to reject the null hypothesis of slope homogeneity. Therefore, having established cross-sectional dependence and slope heterogeneity, all these issues need to be considered in the next steps.

Later, we perform the unit root test that take into account cross-sectional dependence. Thus, we use the unit root tests for panel data of Pesaran (2007). Table 5 represents the results from both, the CIPS and CADF tests. According to panel unit root tests results, for all the variables, the null hypotheses of a unit root cannot be rejected at their levels. However, the variables are stationary in first difference I (1). Therefore, all series are integrated of order one.

Having established the presence of unit roots and that the variables are stationary in first-difference, the next step is to test for the existence of a long-run relationship between variables. Hence, we apply the error-correction-based co-integration tests developed by Westerlund (2007). The results are reported in Table 6. Thus, the p-value allows us confirm a long-run co-integrating relationship between tourism and the informal economy.

Table 5. Results of the second-generation unit root test.

		Variables				
Group	Test	IEC	TI	logGDP	Tax	RL
Levels						
Global	CIPS	−0.649	−0.777	−0.532	−0.622	−0.071
	CADF	−0.805	−0.008	−0.549	−0.162	0.085
HICs	CIPS	−0.492	−0.671	−0.709	−0.865	−0.145
	CADF	−0.828	−1.022	−1.57	−0.849	1.547
HMICs	CIPS	−1.778	−0.616	−0.967	−1.56	−0.347
	CADF	−0.674	−1.126	−0.052	−0.221	−0.403
LMICs	CIPS	−0.76	−0.819	−0.731	−0.949	−0.334
	CADF	−0.848	−0.051	−0.788	−0.615	0.483
LICs	CIPS	−0.191	−1.536	−0.421	−0.46	1.747
	CADF	−0.133	−1.685	−1.084	−0.307	1.547
First differences						
Global	CIPS	−3.786***	−3.914***	−2.669***	−3.759***	−2.208**
	CADF	−1.942**	−3.145***	−2.686***	−3.299***	7.948***
HICs	CIPS	−3.629***	−3.808***	−2.846***	−4.002***	−4.282***
	CADF	−1.965**	−1.159	−1.707**	−2.986***	3.410***
HMICs	CIPS	−3.915***	−3.753***	−3.104***	−3.697	−3.484***
	CADF	−1.811	−3.263***	−3.189***	−2.358***	−3.540***
LMICs	CIPS	−3.897***	−3.956***	−2.868***	−4.086***	−2.471**
	CADF	−1.985**	−2.188**	−1.925	−3.752***	4.346***
LICs	CIPS	−4.327***	−5.673***	−3.558***	−2.597**	2.610***
	CADF	−2.270**	−2.822**	−2.221**	−2.444**	2.810***

The asterisks in the parameters mean: *p < 0.05, **p < 0.01, ***p < 0.001.

Table 6. Result of the Westerlund (2007) co-integration test.

		TI		logGDP		Tax		RL	
Group	Estadístic	Valor	*p*-value	Valor	*p*-value	Valor	*p*-value	Valor	*p*-value
Global	Gt	−3.193	0.000	−3.247	0.000	−4.547	0.000	−4.557	0.000
	Ga	−22.144	0.000	−21.997	0.000	−1.997	0.000	−56.543	0.000
	Pt	−43.352	0.000	−42.875	0.000	−2.764	0.000	−44.175	0.000
	Pa	−24.758	0.000	−24.516	0.000	−55.32	0.000	−4.516	0.000
HICs	Gt	−4.932	0.000	−5.021	0.000	−5.021	0.000	−5.021	0.000
	Ga	−35.455	0.000	−37.795	0.000	−33.543	0.000	−3.795	0.000
	Pt	−30.245	0.000	−31.278	0.000	−11.323	0.000	−1.278	0.000
	Pa	−33.780	0.000	−34.629	0.000	−54.234	0.000	−3.122	0.000
UMICs	Gt	−4.786	0.000	−4.708	0.000	−3.528	0.000	−12.876	0.000
	Ga	−34.909	0.000	−35.280	0.000	−20.252	0.000	−14.322	0.000
	Pt	−25.668	0.000	−25.633	0.000	−23.455	0.000	−5.987	0.000
	Pa	−32.901	0.000	−35.021	0.000	−4.231	0.000	−5.436	0.000
LMICs	Gt	−5.335	0.000	−4.876	0.000	−8.423	0.000	−7.097	0.000
	Ga	−42.906	0.000	−37.048	0.000	−77.043	0.000	−6.867	0.000
	Pt	−28.101	0.000	−25.102	0.000	−24.434	0.000	−21.766	0.000
	Pa	−44.319	0.000	−41.608	0.000	−33.675	0.000	−12.808	0.000
LICs	Gt	−5.381	0.000	−5.202	0.000	−4.665	0.000	−17.302	0.000
	Ga	−40.876	0.000	−40.982	0.000	−1.243	0.000	−65.992	0.000
	Pt	−16.055	0.000	−15.335	0.000	−3.351	0.000	−89.385	0.000
	Pa	−35.202	0.000	−34.258	0.000	−12.234	0.000	−4.432	0.000

The *p*-value was obtained through the bootstrapping method with 200 repetitions.

Once the presence of long-run co-integration among the model variables has been confirmed, we analyze the long-run coefficients of the model by using an FMOLS estimator. First, we calculate the FMOLS coefficients for the whole sample and country group level. Table 7 shows that tourism has a negative but significant impact on the informal economy at the global level and for HICs, UMICs, and LMICs, whereas that tourism increases the informal economy in LICs.

The negative relationship found between tourism and informal economy for HICs, UMICs and LMICs are in line with the theory of dual development or Big Pushes (Banerjee and Duflo, 2011; La Porta and Shleifer, 2014; Lewis, 1954; Rostow, 1990). According to this view, large flows of tourism activity are expected to generate opportunities for the creation of formal firms, operated by entrepreneurs with better human capital and management skills (Hall, 2007; Truong, 2013, 2014; Alam and Paramati, 2016). On the other hand, there is a positive impact of tourism activity on the informal economy for low-income countries, reinforcing the third perspective of informality (Structuralist perspective), which suggests that a greater influx of tourism activity may be accompanied by an increase in the size of the informal economy, possibly due to the fact that several multinational tourism companies based in poor countries and local tourism firms that are part of tourism global production networks hire salaried workers under informal employment relationships (Reid, 2003). In addition, previous work has found linkages between large tourism firms and small local firms, through outsourcing, subcontracting and other types of informal arrangements (Ashley, 2006; Kirsten and Rogerson, 2002; Mshenga and Richardson, 2013).

Table 7. Results of the aggregated FMOLS and DOLS estimators.

	FMOLS					DOLS				
	GLOBAL	HICs	UMICs	LMICs	LICs	GLOBAL	HICs	UMICs	LMICs	LICs
TI	−0.04***	−0.06***	−0.07**	−0.07***	0.02***	−0.023***	−0.031***	−0.11**	−0.16***	0.098***
	(−27.33)	(−79.72)	(−119.95)	(−73.73)	(3.89)	(−4.66)	(−33.44)	(−17.68)	(−22.11)	(2.11)
logGDP	−0.012***	−0.001**	−0.08**	0.01***	0.14***	−0.067***	−0.013**	−0.14**	0.076***	0.22***
	(−13.22)	(−35.61)	(−161.45)	(87.41)	(5.55)	(−10.11)	(−22.06)	(−69.51)	(7.71)	(6.21)
TAX	−0.12***	−0.06***	−0.31**	0.11***	0.17*	−0.19***	−0.12***	−0.24**	0.21***	0.11
	(−12.11)	(−11.75)	(−34.71)	(8.66)	(2.00)	(−10.03)	(−5.51)	(−22.22)	(5.33)	(1.10)
RL	−0.19**	−0.11**	−0.04***	−0.17***	−0.40***	−0.11**	−0.21**	−0.10***	−0.09***	−0.38***
	(−41.33)	(−122.42)	(−32.00)	(−46.87)	(−18.34)	(−36.12)	(−98.35)	(−24.23)	(−35.66)	(−12.12)
Observations	1596	861	273	420	42	1596	861	273	420	42
N	76	41	13	20	2	76	41	13	20	2
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

The asterisks in the parameters mean: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. *t*-statistic in parenthesis.

Table 8. Results from co-integration coefficients FMOLS long-run analysis for the HICs.

	Dependent variable: Informal economy							
Country	TI	*t*-stat	Tax	*t*-stat	LogGDP	*t*-stat	RL	*t*-stat
Australia	−0.03	−6.76	−0.06	−2.14	0.01	1.55	−0.04	−11.52
Austria	0.02	5.37	−0.05	−0.95	−0.07	−11.26	−0.12	−12.38
Bahamas	−0.59	−28.86	0.53	8.87	0.20	25.63	0.29	37.18
Belgium	−0.06	−8.36	0.09	0.73	0.05	5.04	0.03	3.92
Canada	−0.10	−24.78	−0.36	−8.47	0.07	18.98	−0.04	−23.59
Chile	−0.08	−6.63	−0.39	−4.58	0.06	5.24	0.02	1.89
Croatia	−0.24	−45.78	0.37	3.95	0.22	41.94	−0.07	−21.13
Cyprus	0.05	11.75	−0.13	−16.99	−0.02	−1.80	0.05	3.41
Czech Republic	−0.09	−36.68	0.30	3.28	0.06	24.75	0.02	11.24
Denmark	0.05	19.63	−0.24	−8.12	−0.40	−25.80	−0.71	−27.07
Estonia	−0.02	−9.93	0.71	19.07	−0.15	−35.84	0.40	51.07
Finland	−0.01	9.40	−0.25	−10.99	−0.13	−51.26	0.25	61.45
France	−0.07	−38.67	−0.47	−11.29	0.08	36.79	−0.00	−0.12
Germany	−0.07	−12.44	−1.52	−6.83	0.08	15.32	−0.01	−5.83
Greece	−0.07	−36.54	1.03	38.39	0.05	48.44	0.00	2.02
Hungary	−0.11	−19.67	0.54	10.71	0.08	22.50	0.02	25.95
Iceland	−0.01	−14.00	−0.08	−20.36	−0.04	−19.84	0.10	32.47
Ireland	−0.01	−4.12	0.32	12.47	−0.05	−9.85	0.11	13.92
Israel	−0.01	−9.52	0.01	0.57	−0.13	−38.53	0.32	49.95
Italy	−0.01	−1.35	0.47	4.60	0.02	3.93	−0.00	−3.14
Japan	−0.04	−6.32	−0.35	−2.49	0.05	7.95	−0.01	−1.74
Korea, Rep	−0.13	−122.04	0.05	1.17	0.13	135.33	0.00	0.67
Kuwait	0.02	4.03	4.38	4.29	−0.00	−0.59	−0.01	−2.01
Latvia	0.04	11.61	−0.50	−21.24	−0.22	−30.33	−0.43	−43.70
Lithuania	0.01	2.15	−0.08	−3.11	−0.18	−26.70	0.47	42.33
Luxembourg	0.03	126.46	−0.01	−1.12	−0.10	−176.72	0.18	200.10
Malta	−0.04	−7.02	0.01	1.91	0.04	14.59	−0.01	−5.61
Netherlands	−0.02	−1.75	0.10	0.51	−0.09	−5.34	0.21	8.44
New Zealand	−0.03	−26.05	0.27	27.66	0.03	17.80	−0.01	−5.80
Norway	0.00	0.26	−0.34	−6.68	−0.04	−1.36	0.11	2.26
Poland	−0.11	−8.55	2.83	32.28	0.09	11.36	−0.05	−6.54
Portugal	−0.03	−13.97	−0.46	−7.96	0.05	23.71	0.01	6.16
Singapore	−0.03	−25.13	−0.14	−5.20	0.02	25.17	0.02	16.31
Slovak Rep.	−0.06	−38.27	0.19	4.01	0.05	33.13	0.00	1.18
Slovenia	−0.26	−26.66	−2.29	−24.32	0.27	25.17	−0.12	−15.20
Spain	−0.05	−34.46	−0.19	−10.18	0.06	45.79	0.01	27.61
Sweden	−0.04	8.29	−0.22	−9.81	−0.23	−19.06	0.39	25.09
Switzerland	−0.01	−26.94	−0.73	−35.65	0.02	50.95	0.00	3.60
Trin.and Tobago	−0.02	−1.57	−0.30	−4.77	0.03	5.89	0.05	30.23
UK	0.01	0.72	−0.28	−4.40	−0.00	−0.01	0.02	7.29
United States	−0.01	−12.07	−0.44	−24.07	0.02	17.41	0.01	15.10
Uruguay	−0.21	−61.39	0.15	1.10	0.09	20.71	0.21	34.50
Panel results	−0.06	−79.72	−0.06	−11.75	−0.00	−35.61	−0.11	−122.4

Table 9. Results from co-integration coefficients FMOLS long-run analysis for UMICs.

	Dependent variable: informal economy							
Country	TI	*t*-stat	TAX	*t*-stat	LogGDP	*t*-stat	RL	*t*-stat
Argentina	−0.06	−26.34	−0.80	−14.17	0.08	42.68	−0.00	−6.43
Belarus	0.15	16.34	0.00	0.02	−0.30	−20.32	0.60	27.42
Brazil	−0.07	−41.44	0.05	1.15	0.09	50.46	0.00	1.99
Bulgaria	−0.00	−0.21	0.04	0.15	0.01	1.07	0.06	12.47
Costa Rica	−0.08	−41.29	−0.39	−7.27	0.08	75.26	−0.01	−5.49
Dominican Rep.	−0.20	−39.07	0.51	8.27	0.17	44.42	−0.01	−8.97
Guatemala	−0.08	−8.18	−0.18	−0.32	0.12	12.70	−0.02	−1.65
Iran, Islamic Rep.	−0.01	−4.49	0.19	4.49	0.01	4.53	0.02	6.78
Jamaica	−0.12	−4.54	0.05	0.13	0.12	14.16	−0.02	−4.39
Jordan	−0.06	−61.01	−0.39	−31.11	0.06	35.91	0.03	15.50
Kazakhstan	−0.08	−5.90	−0.87	−14.05	0.06	5.69	0.09	9.56
Malaysia	−0.06	−50.32	−0.31	−11.73	0.08	69.38	0.00	3.24
Mexico	−0.03	−8.72	0.39	4.65	0.05	20.87	0.01	16.20
Namibia	0.04	2.76	0.50	5.34	−0.05	−4.68	0.06	14.66
Peru	−0.21	−60.01	−1.52	−12.72	0.24	60.49	−0.04	−5.47
Romania	−0.10	−42.40	−0.18	−2.19	0.09	31.39	0.02	13.96
Russia	−0.23	−37.54	−1.89	−25.17	0.26	43.41	0.02	6.16
South Africa	−0.09	−46.02	−0.87	−20.18	0.12	99.05	−0.00	−0.84
Sri Lanka	−0.10	−5.80	0.45	0.88	0.11	7.20	0.00	0.08
Thailand	−0.08	−70.93	−0.92	−42.12	0.15	127.73	0.01	42.62
Panel results	−0.07	−119.95	−0.31	−34.71	−0.08	−161.45	−0.04	−32.0

Table 10. Results from co-integration coefficients FMOLS long-run analysis for LMICs.

	Dependent variable: Informal economy							
Country	TI	*t*-stat	TAX	*t*-stat	LogGDP	*t*-stat	RL	*t*-stat
Angola	−0.08	−19.64	−0.26	−3.30	0.09	16.96	0.02	1.74
Egypt	−0.09	−29.15	−0.19	−3.16	0.13	39.19	−0.00	−1.27
El Salvador	−0.05	−23.82	−0.52	−11.84	0.09	103.12	−0.00	−3.40
India	−0.08	−8.67	−0.33	−1.36	0.09	4.25	0.04	1.80
Indonesia	−0.10	−47.15	−0.26	−13.30	0.12	58.69	−0.01	−16.20
Mongolia	−0.01	−4.14	0.03	2.68	−0.06	−20.85	0.16	34.01
Morocco	−0.04	−4.91	0.14	1.57	0.05	5.21	0.04	9.17
Myanmar	−0.01	−0.92	3.28	5.16	0.03	3.86	0.07	8.21
Nicaragua	−0.06	−14.57	0.43	4.24	0.07	24.03	0.03	7.20
Philippines	−0.15	−127.77	−0.58	−13.88	0.18	146.05	0.03	30.70
Tunisia	−0.30	−7.66	0.66	1.42	0.20	4.08	0.06	1.62
Ukraine	0.04	18.19	0.24	9.45	−0.40	−48.00	0.81	66.10
Zambia	0.03	4.36	−1.19	−8.91	−0.46	−21.42	0.94	29.27
Panel results	−0.07	−73.73	0.11	−8.66	0.01	87.41	−0.17	−46.87

Also, we performed a robustness check of the previous results. On the right hand-side of the Table 7, we show the dynamic ordinary least squares (DOLS) models. These results confirm the findings we have previously discussed, that is, there is a negative impact of tourism on the informal economy at the global level and for HICs, UMICs, and LMICs, except LICs. In Table B6 (Appendix B), we also show that the results hold when we use the tourism activity index developed with principal component analysis.

While there is a negative impact of tourism on the informal economy in the group of HICs, UMICs, and LMICs, analyzing this relationship at the country level, we find a positive effect of tourism on the informal economy for the following countries: Australia, Cyprus, Denmark, Kuwait, Latvia, Lithuania, and Luxembourg in the group of high-income countries (Table 8); Belarus and Namibia in upper-middle income countries (Table 9); and Ukraine and Zambia in lower-middle income countries (Table 10). For the group of low-income countries, the positive overall relationship remains at the country level (Table 11). This shows the relevance of conducting the analysis at the country level to identify these heterogeneous effects.

Finally, in the last step of this research and after estimating the long-term FMOLS coefficients, we applied the method of Dumitrescu and Hurlin (2012) to investigate the causal relationships between the variables. The results of the Dumitrescu and Hurlin (2012) heterogeneous panel causality test are illustrated in Table 12. The most interesting finding for our variables of interest suggest that there is a bidirectional causal relationship between tourism and the informal economy for the global, upper-middle income, and lower-middle income panels, and a unidirectional causal relationship running from tourism to the informal economy for low-income countries and from the informal economy to tourism for high-income countries. This indicates that public policies focused on tourism activity could have a direct impact on the informal economy. In addition, Table A1 (Appendix A) shows the causality tests between independent variables.

As a comparison, we also conduct the panel VECM test based on Pooled Mean Group (PMG) estimator (Pesaran et al., 1999). Also, this test allows to account for the long-run and short-run dynamic linkages. The results from panel VECM confirm in the short-run and long-run the findings obtained from Dumitrescu-Hurlin Granger causality test for our variables of interest. Results from panel VECM test are reported in Table 13.

As a verification of the robustness of our results, we check whether the results are driven by the choice of the method for calculating the index of tourism activity. As explained earlier, we employ the principal component analysis method to build a weighted tourism index of all the three tourism indicators. The results obtained with the weighted tourism index are consistent with those obtained with the original index. Therefore, we can conclude that our results are robust in regards to specification changes (see Tables B2–B13 in Appendix B).

Table 11. Results from co-integration coefficients FMOLS long-run analysis for the LICs.

	Dependent variable: Informal economy							
Country	TI	t-stat	TAX	t-stat	LogGDP	t-stat	LAO	t-stat
Ethiopia	0.02	2.39	−0.35	−2.92	0.03	3.26	0.13	19.34
Mali	0.03	3.11	0.02	0.10	0.24	4.59	−0.66	−6.59
Panel results	0.02	3.89	−0.17	−2.00	0.14	5.55	−0.40	−18.34

Table 12. Results of Dumitrescu and Hurlin (2012) panel causality test.

Null hypothesis	Group	Z-bar	p-value	Result
TI ≠ > IEC	Global	1.9955*	0.0460	Causality relationship
	HICs	10.947	0.2736	No causality relationship
	UMICs	2.5029*	0.0123	Causality relationship
	LMICs	−11.293*	0.0288	Causality relationship
	LICs	3.0903**	0.0020	Causality relationship
IEC ≠ > TI	Global	3.4844***	0.0005	Causality relationship
	HICs	1.9856*	0.0471	Causality relationship
	UMICs	3.9183***	0.0001	Causality relationship
	LMICs	−0.1453	0.8844	No causality relationship
	LICs	12.555	0.2093	No causality relationship
logGDP ≠ > IEC	Global	8.1429***	0.0000	Causality relationship
	HICs	8.9692***	0.0000	Causality relationship
	UMICs	3.2538***	0.0011	Causality relationship
	LMICs	−0.0440	0.9649	No causality relationship
	LICs	0.4477	0.6544	No causality relationship
IEC ≠ > logGDP	Global	6.9230***	0.0000	Causality relationship
	HICs	8.1847***	0.0000	Causality relationship
	UMICs	0.8221	0.4110	No causality relationship
	LMICs	16.386	0.1013	No causality relationship
	LICs	−10.279	0.3040	No causality relationship
TAX ≠ > IEC	Global	7.4937***	0.0000	Causality relationship
	HICs	6.2288***	0.0000	Causality relationship
	UMICs	4.3819***	0.0000	Causality relationship
	LMICs	2.1587*	0.0309	Causality relationship
	LICs	−0.9129	0.3613	No causality relationship
IEC ≠ > TAX	Global	3.4133***	0.0006	Causality relationship
	HICs	4.2066***	0.0000	Causality relationship
	UMICs	0.0752	0.9401	No causality relationship
	LMICs	0.5638	0.5729	No causality relationship
	LICs	0.4083	0.6830	No causality relationship
RL ≠ > IEC	Global	0.001	0.98	No causality relationship
	HICs	0.001	0.98	No causality relationship
	UMICs	17.903	0.0734	No causality relationship
	LMICs	0.001	0.98	No causality relationship
	LICs	0.001	0.98	No causality relationship
IEC ≠ > RL	Global	0.001	0.98	No causality relationship
	HICs	0.001	0.98	No causality relationship
	UMICs	2.7338**	0.0063	Causality relationship
	LMICs	0.001	0.98	No causality relationship
	LICs	0.001	0.98	No causality relationship

The asterisks in the parameters mean: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. ≠ > means variable x does not Granger cause variable.

Table 13. Result of the panel Granger causality with dynamic vector error-correction model (VECM).

Group	Dependent variable	Short-run relationship					Long-run relationship
		IEC	TI	LogGDP	TAX		ECT
GLOBAL	IEC	–	−0.002***(0.000)	−0.032***(0.000)	−0.179**(0.010)	0.001***(0.000)	−0.275**(0.002)
	TI	−1.06***(0.000)	–	1.019***(0.000)	0.771***(0.000)	−0.008 (0.19)	−0.192***(0.000)
	LogGDP	−1.10***(0.000)	0.073***(0.000)	–	0.120***(0.000)	0.005***(0.000)	−0955***(0.000)
	TAX	−0.09***(0.000)	0.015***(0.000)	0.072***(0.000)	–	−0.001***(0.000)	−0.061***(0.000)
	RL	−0.21 (0.300)	0.066***(0.000)	0.094***(0.000)	0.091***(0.000)	–	−0.005* (0.025)
HICs	IEC	–	−0.003 (0.128)	−0.028***(0.000)	−0.29** (0.009)	0.022 (0.320)	−0.514***(0.000)
	TI	−1.11** (0.030)	–	0.026***(0.000)	0.444***(0.000)	−0.013**(0.029)	−0.192* (0.042)
	LogGDP	−1.87***(0.000)	0.043 (0.062)	–	0.220 (0.200)	0.003***(0.000)	0642***(0.000)
	TAX	−0.06***(0.000)	0.021***(0.000)	0.011***(0.000)	–	−0.002 (0.210)	−0.034***(0.000)
	RL	−0.07***(0.000)	0.068***(0.000)	0.043***(0.000)	0.068 (0.000)	–	−0.024* (0.032)
UMICs	IEC	-	−0.004***(0.000)	−0.016*(0.032)	−0.144***(0.000)	0.051* (0.040)	−0.314***(0.000)
	TI	−1.17**(0.010)	–	1.016***(0.000)	0.325 (0.414)	−0.012 (0.16)	−0.162***(0.000)
	LogGDP	−1.66 (0.600)	0.042***(0.000)	-	0.120 (0.000)	0.005 (0.740)	0.611***(0.000)
	TAX	−0.034*(0.031)	0.015 (0.241)	0.044 (1.210)	-	−0.001***(0.000)	−0.056 (0.056)
	RL	−0.06***(0.000)	0.017***(0.000)	0.069***(0.000)	0.064 (0.000)	-	0.114***(0.000)
LMICs	IEC	–	−0.002***(0.001)	−0.021 (0.508)	−0.152***(0.000)	0.003 (0.300)	−0.114***(0.000)
	TI	−1.22 (0.110)	–	1.229***(0.000)	0.544 (1.300)	−0.014 (0.12)	−0.152***(0.000)
	LogGDP	−1.240 (0.090)	0.044***(0.000)	–	0.130* (0.020)	0.006***(0.000)	−0948 (0.550)
	TAX	−0.88 (0.900)	0.025 2 (0.067)	0.033 (0.210)	-	−0.001 (0.900)	−0.069** (0.020)
	RL	−0.21 (0.300)	0.041***(0.000)	0.042***(0.000)	0.011***(0.000)	-	0.334***(0.000)
LICs	IEC	–	−0.003*(0.031)	−0.021 (0.099)	−0.159 (0.069)	0.005 (0.320)	−0.444***(0.000)
	TI	1.06 (0.057)	–	1.441 (0.230)	0.773 (0.920)	−0.008***(0.000)	0.556* (0.018)
	LogGDP	−0.77 (0.968)	0.081***(0.000)	–	0.120* (0.034)	0.007 (0.210)	−0223***(0.000)
	TAX	−0.04 (0.080)	0.015* (0.021)	0.002***(0.000)	–	−0.021***(0.000)	0.554* (0.031)
	RL	−0.81 (0.730)	0.015 (0.310)	0.091***(0.000)	0.072 (0.000)	-	−0.064 (0.066)

The asterisks in the parameters mean: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. p-value in parenthesis.

Conclusions and policy implications

In this study, the long-run and causal relationship between tourism and the informal economy was investigated, both globally, considering 76 countries and by groups of countries classified according to the Atlas method for the period 1995–2015. To the best of our knowledge, fewer studies have focused on the relationship between tourism and informal economy, except Lv (2020). However, it is for the first time to be considered the long-run relationship between tourism and the informal economy.

After detecting the existence of cross-sectional dependence and slope heterogeneity, we employed the second-generation panel unit root test of Pesaran (2007) and the co-integration test developed by Westerlund (2007). The results show that all series are stationary at first differences, and a long-run relationship between tourism and informal economy is confirmed. Then, a fully modified ordinary least squares (FMOLS) estimator (Pedroni, 2001) was employed to determine the co-integrating coefficients. The results show that tourism has a negative but significant impact on the informal economy at the global level and for HICs, UMICs, and LMICs, whereas that tourism increases the informal economy in low-income countries. Finally, Dumitrescu-Hurlin panel causality test, shows evidence of significant bidirectional causality relationship between tourism and the informal economy for the global, UMICs, and LMICs panels. Also, we note the presence of unidirectional causality running from tourism to the informal economy for LICs and from the informal economy to tourism for HICs.

The results support the existence of a relationship between tourism and the informal economy that varies as countries advance in their level of development. For instance, for low-income countries, we find a positive relationship in the short and long-run between tourism and the informal economy. Then, for low-income countries, the expansion of tourism has generated employment and entrepreneurship opportunities, especially in the informal sector. Probably, there are links between large tourism enterprises and small local enterprises, through outsourcing, subcontracting and other types of informal arrangements, as already warned by some scholars of the structuralist view of informality. On the other hand, in the case of HICs, UMICs, and LMICs, the increase in tourism activity has been beneficial for reducing the size of the informal economy. These results support the dual view of informality. That is, tourism is a key sector for promoting economic growth, and thus a relevant instrument of poverty alleviation and modernization of the economy, as suggested by some modernization theorists. Our findings suggest that, for these countries, large flows of tourism activity have generated opportunities for the creation of formal firms.

From a public policy point of view, countries in each of these stages of development require different types of public policy. For instance, low-income countries should design tourism incentive policies that promote formal tourism ventures operated by entrepreneurs with better human capital and management skills, and additionally assist and motivate informal tourism entrepreneurs to formalize their activities. Additionally, policy-makers in low-income countries should relax some of the restrictions that many tourism informal entrepreneurs face, such as capital and retail space, encouraging informal entrepreneurs to formalize their activities by increasing the benefits of operating formally. These actions can significantly benefit the communities where tourist attractions are located. We are able to derive this hypothesis taking in consideration that panel causality tests show a unidirectional relationship running from tourism to the informal economy in low-income countries. On the other hand, taking into account the negative relationship between tourism and the informal economy for HICs, UMICs, and LMICs, these countries could reduce their levels of informal economy, not only by focusing on traditional causes of informality, such as excessive regulation or tax burden, but also by strategically stimulating tourism activity.

It is important to indicate that our findings are consistent with the negative and significant relationship between tourism and the informal economy found by Lv (2020) for 96 countries over the time period from 2000 to 2007. Nonetheless, we make a clear methodological contribution derived from the use of co-integration techniques. Indeed, to the best of our knowledge, we are the first to use these techniques to investigate this relationship with the corresponding benefits that we describe next. First, the use of this technique allow for a dynamic analysis that enables us to treat all variables as endogenous (Berdiev and Saunoris, 2016). Furthermore, this dynamic analysis allows us to capture changes of the investigated relationship in the long-run that were not considered in the previous study by Lv (2020). Besides, our approach also allows us to test the causal relationships between the variables. Finally, the technique allows us to perform the analysis at the level of countries.

Finally, the results for low-income countries should be interpreted with caution, as they include a small number of countries. Future research could attempt to expand the sample of low-income countries. In addition, our study identify heterogeneous results at the country level, Therefore, exploring case studies for specific countries would be a very interesting exercise and would complement the findings of the present study. Likewise, other studies could use other proxies to measure the size of the informal economy, and even better if it is feasible to obtain data on the informal economy at the level of tourism activities.

Acknowledgments

The authors gratefully acknowledge the wise comments of two anonymous referees and the support of the research center of the Universidad de Especialidades Espiritu Santo.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

ORCID iDs

Cristian Ortiz https://orcid.org/0000-0002-9395-7228
Pablo Ponce https://orcid.org/0000-0003-2717-0176

References

Adiyia B, Stoffelen A, Jennes B, et al. (2015) Analysing governance in tourism value chains to reshape the tourist bubble in developing countries: the case of cultural tourism in Uganda. *Journal of Ecotourism* 14(2–3): 113–129.

Akaike H (1974) A new look at the statistical model identification. *IEEE Transactions on Automatic Control* 19(6): 716–723.

Alam MS and Paramati SR (2016) The impact of tourism on income inequality in developing economies: does kuznets curve hypothesis exist? *Annals of Tourism Research* 61: 111–126.

Alrawadieh Z and Alrawadieh Z (2018) Exploring entrepreneurship in the sharing accommodation sector: empirical evidence from a developing country. *Tourism Management Perspectives* 28: 179–188.

Alvarado R, Ortiz C, Jiménez N, et al. (2020) Ecological footprint, air quality and research and development: the role of agriculture and international trade. *Journal of Cleaner Production*: 125589.

Anderson W and Juma S (2011) Linkages at tourism destinations: challenges in Zanzibar. *ARA Journal of Tourism Research* 3(1): 27–41.

Ashley C (2006) *Participation by the poor in Luang Prabang tourism economy: current earnings and opportunities for expansion*. London: ODI.

Autio E and Fu K (2015) Economic and political institutions and entry into formal and informal entrepreneurship. *Asia Pacific Journal of Management* 32(1): 67–94.

Balsalobre-Lorente D, Driha OM and Sinha A (2020) The dynamic effects of globalization process in analysing n-shaped tourism led growth hypothesis. *Journal of Hospitality and Tourism Management* 43: 42–52.

Banerjee AV and Duflo E (2011) *Poor economics: rethinking poverty & the ways to end it*. India: Random House.

Baklouti N and Boujelbene Y (2020) A simultaneous equation model of economic growth and shadow economy: is there a difference between the developed and developing countries? *Economic Change and Restructuring* 53(1): 151–170.

Baum T and Hai NTT (2019) Applying sustainable employment principles in the tourism industry: righting human rights wrongs? *Tourism Recreation Research* 44(3): 371–381.

Berdiev AN and Saunoris JW (2016) Financial development and the shadow economy: a panel VAR analysis. *Economic Modelling* 57: 197–207.

Bianchi RV and de Man F (2021) Tourism, inclusive growth and decent work: a political economy critique. *Journal of Sustainable Tourism* 29(2–3): 353–371.

Boonjubun C (2017) Conflicts over streets: the eviction of bangkok street vendors. *Cities* 70: 22–31.

Breitung J and Das S (2005) Panel unit root tests under cross-sectional dependence. *Statistica Neerlandica* 59(4): 414–433.

Çakmak E, Lie R and McCabe S (2018) Reframing informal tourism entrepreneurial practices: capital and field relations structuring the informal tourism economy of chiang mai. *Annals of Tourism Research* 72: 37–47.

Castells M and Portes A (1989) World underneath: the origins, dynamics, and effects of the informal economy. In: Portes A, Castells M and Benton L (eds), *The informal economy: studies in advanced and less developed countries*. Baltimore: John Hopkins University Press.

Çenesiz A and Çakmak E (2020) Measuring the size of the informal tourism economy in Thailand. *International Journal of Tourism Research* 22: 37–47.

Chang S-C (2015) Effects of financial developments and income on energy consumption. *International Review of Economics & Finance* 35: 28–44.

Chavarria LC and Phakdee-auksorn P (2017) Understanding international tourists' attitudes towards street food in Phuket, Thailand. *Tourism Management Perspectives* 21: 66–73.

Chen MA (2006) Rethinking the informal economy: linkages with the formal economy and the formal regulatory environment. In: Guha-Khasnobis B, Kanbur R and Ostrom E (eds.) *Linking the Formal and Informal Economy – Concepts and Policies*. Oxford: Oxford University Press, 75–92.

Chrid N, Saafi S and Chakroun M (2020) Export upgrading and economic growth: a panel cointegration and causality analysis. *Journal of the Knowledge Economy*: 1–31.

Clancy M (2011) Global commodity chains and tourism: past Research and future directions. In: Mosedale J (eds), *Political economy of tourism: a critical perspective*. London: Routledge, 75–92.

Coe NM and Yeung HWC (2015) *Global production networks: theorizing economic development in an interconnected world*. Oxford University Press.

Colombo E, Menna L and Tirelli P (2019) Informality and the labor market effects of financial crises. *World Development* 119: 1–22.

Christian M (2016) Tourism global production networks and uneven social upgrading in Kenya and Uganda. *Tropical Gastroenterology: Official Journal of the Digestive Diseases Foundation* 18(1): 38–58.

Croes R (2014) The role of tourism in poverty reduction: an empirical assessment. *Tourism Economics* 20(2): 207–226.

Damayanti M, Scott N and Ruhanen L (2017) Coopetitive behaviours in an informal tourism economy. *Annals of Tourism Research* 65: 25–35.

Damayanti M, Scott N and Ruhanen L (2018) Space for the informal tourism economy. *The Service Industries Journal* 38(11–12): 772–788.

Dau LA and Cuervo-Cazurra A (2014) To formalize or not to formalize: entrepreneurship and pro-market institutions. *Journal of Business Venturing* 29(5): 668–686.

De Castro JO, Khavul S and Bruton GD (2014) Shades of grey: how do informal firms navigate between macro and meso institutional environments? *Strategic Entrepreneurship Journal* 8(1): 75–94.

De Paula A and Scheinkman JA (2011) The informal sector: an equilibrium model and some empirical evidence from Brazil. *Review of Income and Wealth* 57 :S8–S26.

De Soto H (1989) *The other path: the invisible revolution in the Third World*. New York: Harper and Row publishers.

De Soto H (2000) *The Mystery of Capital: Why Capitalism Triumphs in the West and Fails Everywhere Else*. New York: Basic Books.

Duffy R (2000) Shadow players: ecotourism development, corruption and state politics in Belize. *Third World Quarterly* 21(3): 549–565.

Dumitrescu E-I and Hurlin C (2012) Testing for granger non-causality in heterogeneous panels. *Economic Modelling* 29(4): 1450–1460.

Duncan T, Scott DG and Baum T (2013) The mobilities of hospitality work: an exploration of issues and debates. *Annals of Tourism Research* 41: 1–19.

Elgin C and Oztunali O (2014) Institutions, informal economy, and economic development. *Emerging Markets Finance and Trade* 50(4): 145–162.

Enste DH (2010) Shadow economy – The impact of regulation in OECD-countries. *International Economic Journal* 24(4): 555–571.

Eugenio-Martin JL, Martín Morales N and Scarpa R (2004) Tourism and economic growth in Latin American countries: a panel data approach. *SSRN Electronic Journal*.

Fayissa B, Nsiah C and Tadesse B (2009) Tourism and economic growth in latin american countries (Lac): further empirical evidence. Department of Economics and Finance Working Paper Series. Online available at: http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.587.6092&rep=rep1&type=pdf.

Feige EL (1990) Defining and estimating underground and informal economies: the new institutional economics approach. *World Development* 18(7): 989–1002.

Friedman E, Johnson S, Kaufmann D, et al. (2000) Dodging the grabbing hand: the determinants of unofficial activity in 69 countries. *Journal of Public Economics* 76(3): 459–493.

Gereffi G (1999). *A commodity chains framework for analyzing global industries*. Institute of Development Studies, Vol. 8, pp. 1–9. Online availible at: http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.608.812&rep=rep1&type=pdf

Gerxhani K (2004) The informal sector in developed and less developed countries: a literature survey. *Public Choice* 120(3): 267–300.

Gindling TH and Newhouse D (2014) Self-employment in the developing world. *World Development*. 56: 313–331.

Guha-Khasnobis B, Kanbur R and Elinor O (2006) *Linking the formal and informal economy: concepts and policies*. Oxford: OUP.

Guttentag D (2015) Airbnb: disruptive innovation and the rise of an informal tourism accommodation sector. *Current issues in Tourism* 18(12): 1192–1217.

Gutiérrez-Romero R (2021) Inequality, persistence of the informal economy, and club convergence. *World Development* 139: 105211.

Hall CM (2007) Pro-poor tourism: do 'tourism exchanges benefit primarily the countries of the south'? *Cardiovascular Intervention and Therapeutics* 10(2–3): 111–118.

Helmke G and Levitsky S (2004) Informal institutions and comparative politics: a research agenda. *Perspectives on Politics* 2(4): 725–740.

Henderson JC and Smith RA (2009) The informal tourism economy at beach resorts: a comparison of cha-am and laguna phuket in thailand. *Tourism Recreation Research* 34(1): 13–22.

Huynh CM and Nguyen TL (2020) Fiscal policy and shadow economy in Asian developing countries: does corruption matter? *Empirical Economics* 59(4): 1745–1761.

Janta H, Lugosi P, Brown L, et al. (2012) Migrant networks, language learning and tourism employment. *Tourism Management* 33(2): 431–439.

Johnson S, Kaufmann D and Zoido-Lobaton P (1998) Regulatory discretion and the unofficial economy. *The American Economic Review* 88(2): 387–392.

Jones T, Ram M and Edwards P (2006) *International Journal of Sociology and Social Policy.*Shades of grey in the informal economy

Khan, MTI, Yaseen MR and Ali Q (2019) Nexus between financial development, tourism, renewable energy, and greenhouse gas emission in high-income countries: a continent-wise analysis. *Energy Economics* 83: 293–310.

Kirsten M and Rogerson CM (2002) Tourism, business linkages and small enterprise development in South Africa. *Development Southern Africa* 19(1): 29–59.

Koçak E, Ulucak R and Ulucak ZS (2020) The impact of tourism developments on co2 emissions: an advanced panel data estimation. *Tourism Management Perspectives* 33: 100611.

La Porta R and Shleifer A (2008) The unofficial economy and economic development. *Brookings Papers on Economic Activity* 39(2): 275–363.

La Porta R and Shleifer A (2014) Informality and development. *Journal of Economic Perspectives* 28(3): 109–126.

Lashley C and Rowson B (2010) Lifestyle businesses: insights into Blackpool's hotel sector. *International Journal of Hospitality Management* 29(3): 511–519.

Lewis WA (1954) Economic development with unlimited supplies of labour. *The Manchester School* 22(2): 139–191.

Liddle B (2012) The importance of energy quality in energy intensive manufacturing: evidence from panel cointegration and panel fmols. *Energy Economics* 34(6): 1819–1825.

Liu A and Wall G (2006) Planning tourism employment: a developing country perspective. *Tourism Management* 27(1): 159–170.

Loayza NV (2016) Informality in the Process of Development and Growth. *The World Economy* 39(12): 1856–1916.

Lv Z (2020) Does tourism affect the informal sector? *Annals of Tourism Research* 80: 102816.

Margolis DN (2014) By choice and by necessity: entrepreneurship and self-employment in the developing world. *The European Journal of Development Research* 26(4): 419–436.

Mbaiwa JE and Hambira WL (2020) Enclaves and shadow state tourism in the Okavango Delta, Botswana. *South African Geographical Journal* 102(1): 1–21.

Medina L and Schneider F (2018) Shadow economies around the world: what did we learn over the last 20 years? *IMF Working Papers* 18: 1–76.

Mshenga PM and Richardson RB (2013) Micro and small enterprise participation in tourism in coastal Kenya. *Small Business Economics* 41(3): 667–681.

Mshenga PM, Richardson RB, Njehia BK, et al. (2010) The contribution of tourism to micro and small enterprise growth. *Tourism Economics* 16(4): 953–964.

Murphy JT (2019) Global production network dis/articulations in Zanzibar: practices and conjunctures of exclusionary development in the tourism industry. *Journal of Economic Geography* 19(4): 943–971.

Murphy KM, Shleifer A and Vishny RW (1989) Industrialization and the big push. *Journal of Political Economy* 97(5): 1003–1026.

Nikopour H and Shah Habibullah M (2010) *Shadow economy and poverty.* MPRA Paper.

North DC (1990) *Institutions, institutional change and economic performance*. Cambridge University Press.

Onodugo VA, Ezeadichie NH, Onwuneme CA, et al. (2016) The dilemma of managing the challenges of street vending in public spaces: the case of Enugu City, Nigeria. *Cities* 59: 95–101.

Ortiz C, Alvarado R and Salinas A (2019) The effect of military spending on output: new evidence at the global and country group levels using panel data cointegration techniques. *Economic Analysis and Policy* 62: 402–414.

Ozcan B (2013) The nexus between carbon emissions, energy consumption and economic growth in middle east countries: a panel data analysis. *Energy Policy* 62: 1138–1147.

Loayza N (1999) *The economics of the informal sector: a simple model and some empirical evidence from Latin America*. The World Bank.

Paramati SR, Shahbaz M and Alam MS (2017) Does tourism degrade environmental quality? a comparative study of eastern and western european union. *Transportation Research Part D: Transport and Environment* 50: 1–13.

Pécot M, Gavilanes J and Sáenz De Viteri A (2018) Tales of informality: tourism development in four Ecuadorian beaches. *Tourism Planning & Development* 15(5): 584–599.

Pedroni P (2001). Fully modified OLS for heterogeneous cointegrated panels In: Baltagi BH, Fomby TB and Carter Hill R (eds), *Nonstationary panels, panel cointegration, and dynamic panels*. Bingley: Emerald Group Publishing Limited, 15, pp. 93–130.

Peng B, Song H and Crouch GI (2014) A meta-analysis of international tourism demand forecasting and implications for practice. *Tourism Management* 45: 181–193.

Peng B, Song H, Crouch GI, et al. (2015) A meta-analysis of international tourism demand elasticities. *Journal of Travel Research* 54(5): 611–633.

Perry G and Maloney T (2007) Evaluating active labour market programmes in New Zealand. *International Journal of Manpower* 28(1).

Persyn D and Westerlund J (2008) Error-correction-based cointegration tests for panel data. *The Stata Journal: Promoting Communications on Statistics and Stata* 8(2): 232–241.

Pesaran MH (2004) General diagnostic tests for cross-sectional dependence in panels. *Empirical Economics*. 1–38.

Pesaran MH (2007) A simple panel unit root test in the presence of cross-section dependence. *Journal of Applied Econometrics* 22(2): 265–312.

Pesaran MH (2015) Testing weak cross-sectional dependence in large panels. *Econometric Reviews* 34(6–10): 1089–1117.

Hashem Pesaran M and Yamagata T (2008) Testing slope homogeneity in large panels. *Journal of Econometrics* 142(1): 50–93.

Pesaran MH, Shin Y and Smith RP (1999) Pooled mean group estimation of dynamic heterogeneous panels. *Journal of the American statistical Association* 94: 446–634.

Piore MJ and Sabel CF (1986) *The second industrial divide: possibilities for prosperity*. New York, NY: Basic Books.

Ponce P, Alvarado R, Ponce K, et al. (2019) Green returns of labor income and human capital: empirical evidence of the environmental behavior of households in developing countries. *Ecological economics* 160: 105–113.

Pulido-Fernández JI and Cárdenas-García PJ (2021) Analyzing the bidirectional relationship between tourism growth and economic development. *Journal of Travel Research* 60(3): 583–602.

Reid DG (2003) *Tourism, globalization and development: responsible tourism planning*. London: Pluto Press.

Romero I, Fernández-Serrano J and Cáceres-Carrasco FR (2020) Tour operators and performance of SME hotels: differences between hotels in coastal and inland areas. *International Journal of Hospitality Management* 85: 102348.

Rostow WW (1990) *The Stages of Economic Growth: A Non-communist Manifesto*. New York: Cambridge University Press.

Salinas A, Ortiz C and Muffatto M (2019) Business regulation, rule of law and formal entrepreneurship: evidence from developing countries. *Journal of Entrepreneurship and Public Policy*. 8(2): 254–271.

Salinas A, Ortiz C, Muffatto M, et al. (2020) Formal institutions and informal entrepreneurial activity: panel data evidence from Latin American countries. *Entrepreneurship Research Journal* 1. (ahead-of-print).

Schneider F (2017) Implausible large differences in the sizes of underground economies in highly developed european countries? a comparison of different estimation methods. CESifo Working Paper Series No. 6522. Online available at: https://ssrn.com/abstract=2999761

Schneider F and Buehn A (2013) Estimating the size of the shadow economy: Methods, problems and open questions CESifo Working Paper Series No. 4448. Online availible at: https://www.cesifo.org/DocDL/cesifo1_wp4448.pdf.

Schneider F, Buehn A and Montenegro CE (2010) New estimates for the shadow economies all over the world. *International Economic Journal* 24(4): 443–461.

Schneider F (2015) Size and development of the shadow economy of 31 european and 5 other oecd countries from 2003 to 2014: different developments? *Journal of Self – Governance and Management Economics* 3(4): 7–29.

Sequeira TN and Maçãs Nunes P (2008) Does tourism influence economic growth? A dynamic panel data approach. *Applied Economics* 40(18): 2431–2441.

Snieška V and Bruneckiene J (2009) Measurement of lithuanian regions by regional competitiveness index. *Inžinerinė Ekonomika*(1): 45–57.

Song H, Liu J and Chen G (2013) Tourism value chain governance. *Journal of Travel Research* 52(1): 15–28.

Truong VD (2013) Tourism policy development in Vietnam: a pro-poor perspective. *Journal of Policy Research in Tourism, Leisure and Events* 5(1): 28–45.

Truong VD (2014) Pro-poor tourism: looking backward as we move forward. *Tourism Planning & Development* 11(2): 228–242.

Truong VD (2018) Tourism, poverty alleviation, and the informal economy: the street vendors of Hanoi, Vietnam. *Tourism Recreation Research* 43(1): 52–67.

Wang Q, Kong Y, Zhang W, et al. (2016) Regional debris flow susceptibility analysis based on principal component analysis and self-organizing map: a case study in Southwest China. *Arabian Journal of Geosciences* 9(18): 1–18.

Wang S, Li G and Fang C (2018) Urbanization, economic growth, energy consumption, and CO2 emissions: Empirical evidence from countries with different income levels. *Renewable and Sustainable Energy Reviews* 81: 2144–2159.

Webb JW, Bruton GD, Tihanyi L, et al. (2013) Research on entrepreneurship in the informal economy: framing a research agenda. *Journal of Business Venturing* 28(5): 598–614.

Webb JW, Ireland RD and Ketchen DJ Jr (2014) Toward a greater understanding of entrepreneurship and strategy in the informal economy. *Strategic Entrepreneurship Journal* 8(1): 1–15.

Webb JW, Tihanyi L, Ireland RD, et al. (2009) You say illegal, i say legitimate: entrepreneurship in the informal economy. *Academy of Management Review* 34(3): 492–510.

Welter F and Smallbone D (2011) Institutional perspectives on entrepreneurial behavior in challenging environments. *Journal of Small Business Management* 49(1): 107–125.

Welter F, Smallbone D and Pobol A (2015) Entrepreneurial activity in the informal economy: a missing piece of the entrepreneurship jigsaw puzzle. *Entrepreneurship & Regional Development* 27(5–6): 292–306.

Westerlund J (2007) Testing for error correction in panel data. *Oxford Bulletin of Economics and Statistics* 69(6): 709–748.

Williams CC and Horodnic IA (2017) Regulating the sharing economy to prevent the growth of the informal sector in the hospitality industry. *International Journal of Contemporary Hospitality Management* 29.

Williams CC and Schneider F (2016) *Measuring the global shadow economy*. Number 16551 in Books. Edward Elgar Publishing.

Williams CC, Shahid MS and Martínez A (2016). Determinants of the level of informality of informal microenterprises: some evidence from the city of Lahore, Pakistan. *World Development* 84: 312–325.

Williams N and Vorley T (2014) Economic resilience and entrepreneurship: lessons from the sheffield city region. *Entrepreneurship & Regional Development* 26(3–4): 257–281.

Wooldridge JM (1991) On the application of robust, regression- based diagnostics to models of conditional means and conditional variances. *Journal of Econometrics* 47(1): 5–46.

Wu R and Xie Z (2020) Identifying the impacts of income inequality on CO2 emissions: empirical evidences from OECD countries and non-OECD countries. *Journal of Cleaner Production* 277: 123858.

Xu T and Lv Z (2021) Does too much tourism development really increase the size of the informal economy? *Current Issues in Tourism*: 1–6.

Yeo S-J and Heng CK (2014) An (Extra)ordinary night out: urban informality, social sustainability and the night-time economy. *Urban Studies* 51(4): 712–726.

Zaman K, Shahbaz M, Loganathan N, et al. (2016) Tourism development, energy consumption and Environmental Kuznets Curve: trivariate analysis in the panel of developed and developing countries. *Tourism Management* 54: 275–283.

Author biographies

Aldo Salinas, lecturer and research associate at ESAI Business School, Universidad Espiritu Santo. asalinasa@uees.edu.ec Aldo completed his Ph.D. at Padua University, Italy. His thesis examined the effects of institutional factors on the rate of entrepreneurship in developing countries. His research interest includes informal economic activities, entrepreneurship and Innovation.

Cristian Ortiz, research associate at ESAI Business School, Universidad Espiritu Santo, Guayas, Ecuador. cristian.ortiz@unl.edu.ec Cristian completed his master´s degree at Universidad Catolica del Norte, Chile. His thesis examined individuals' sustainable behaviors and subjective well-being across regions. His research interests include studies in regional and urban economics, environmental economics, and management studies.

Pablo Ponce, lecturer and research associate at Carrera de Economia, Universidad Nacional de Loja, Ecuador. pablo.ponce@unl.edu.ec Pablo completed his master's degree at Universitat de Barcelona, Spain. His thesis examined the effects of a license penalty system on road safety indicators. His

research interest includes applied tourism studies, environmental economics, and the effects of tourism on poverty reduction.

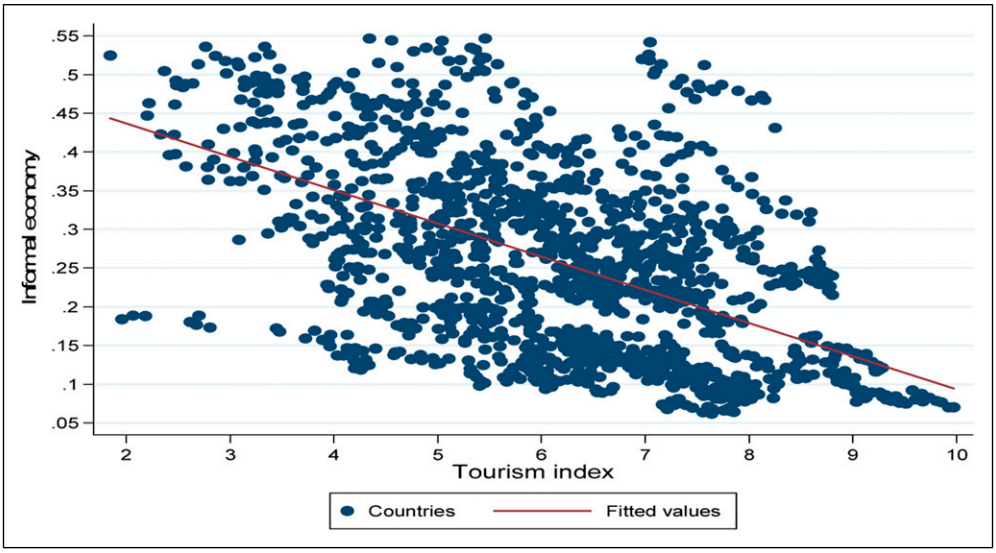


Figure A1. Correlation between informal activity and the tourism index.

Table A1. Results of Dumitrescu and Hurlin (2012) panel causality test between independent variables

Null hypothesis	Group	Z-bar	p-value	Result
TI ≠ > logGDP	Global	2.4411*	0.041	Causality relationship
	HICs	2.447	0.273	No causality relationship
	UMICs	2.1024*	0.022	Causality relationship
	LMICs	−22.243*	0.028	Causality relationship
	LICs	3.0403**	0.002	Causality relationship
logGDP ≠ > TI	Global	3.4844***	0.000	Causality relationship
	HICs	2.4816*	0.047	Causality relationship
	UMICs	3.4283***	0.000	Causality relationship
	LMICs	−0.2413	0.884	No causality relationship
	LICs	22.111***	0.004	Causality relationship
TI ≠ > TAX	Global	8.2424***	0.000	Causality relationship
	HICs	8.4642***	0.000	Causality relationship
	UMICs	3.2138***	0.002	Causality relationship
	LMICs	−0.0440	0.464	No causality relationship
	LICs	0.4477	0.614	No causality relationship
TAX ≠ > TI	Global	6.4230***	0.000	Causality relationship
	HICs	8.2847***	0.000	Causality relationship
	UMICs	0.8222	0.420	No causality relationship
	LMICs	6.386	0.203	No causality relationship
	LICs	−20.274	0.340	No causality relationship

(continued)

Table A1. (continued)

Null hypothesis	Group	Z-bar	p-value	Result
TI ≠ > RL	Global	7.4437***	0.000	Causality relationship
	HICs	6.2288***	0.000	Causality relationship
	UMICs	4.3824***	0.000	Causality relationship
	LMICs	2.2187*	0.030	Causality relationship
	LICs	−0.4224	0.362	No causality relationship
RL ≠ > TI	Global	3.4233***	0.000	Causality relationship
	HICs	4.2066***	0.000	Causality relationship
	UMICs	0.0712	0.440	No causality relationship
	LMICs	0.1638	0.172	No causality relationship
	LICs	0.4083	0.683	No causality relationship
logGDP ≠ > TAX	Global	0.002*	0.048	Causality relationship
	HICs	0.002	0.480	No causality relationship
	UMICs	27.403	0.073	No causality relationship
	LMICs	0.002	0.486	No causality relationship
	LICs	0.042***	0.000	Causality relationship
TAX ≠ > logGDP LogGDP LogGDP	Global	0.012	0.331	No causality relationship
	HICs	0.031	0.482	No causality relationship
	UMICs	2.7338**	0.006	Causality relationship
	LMICs	0.002	0.223	No causality relationship
	LICs	0.002	0.483	No causality relationship
logGDP ≠ > RL	Global	7.4437***	0.000	Causality relationship
	HICs	6.2288***	0.000	Causality relationship
	UMICs	4.3824***	0.000	Causality relationship
	LMICs	2.2187*	0.030	Causality relationship
	LICs	−0.4224	0.362	No causality relationship
RL ≠ > logGDP	Global	3.4233***	0.000	Causality relationship
	HICs	4.2066***	0.000	Causality relationship
	UMICs	0.0712	0.440	No causality relationship
	LMICs	0.1638**	0.024	Causality relationship
	LICs	0.4083	0.683	No causality relationship
RL ≠ > TAX	Global	0.002**	0.023	Causality relationship
	HICs	0.002	0.484	No causality relationship
	UMICs	7.403**	0.003	Causality relationship
	LMICs	1.002	0.481	No causality relationship
	LICs	0.002***	0.001	Causality relationship
TAX ≠ > RL logGDP	Global	0.002	0.482	No causality relationship
	HICs	0.002	0.323	No causality relationship
	UMICs	2.7338**	0.006	Causality relationship
	LMICs	0.002	0.481	No causality relationship
	LICs	0.002	0.338	No causality relationship

The asterisks mean: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. ≠ > means variable x does not Granger cause variable.

Javier Changoluisa, lecturer and research associate at ESAI Business School, Universidad Espiritu Santo. Jchangoluisa@uees.edu.ec Javier completed his Ph.D. at Friedrich Schiller University in

Table B1. Principal component analysis for weighted tourism index.

Panel A: Eigenvalues of the observed matrix				
Component	Eigenvalue	Difference	Proportion	Cumulative
Comp1	2.70813	2.48736	0.9027	0.9027
Comp2	0.22077	0.14967	0.0736	0.9763
Comp3	0.07110		0.0237	1.0000
Panel B: Eigenvectors (loadings)				
Variable	Comp1	Comp2	Comp3	
TE	0.6031	−0.4071	0.6071	
TA	0.59736	−0.467	−0.7361	
TR	0.5737	0.376	−0.5237	
Panel C: Reliability and validity test			Tourism activity index	
Number of items			3	
Average inter-item correlation			0.8535	
Cronbach's alpha			0.9454	
Kayser Meyer Olin measure (KMO)			0.881	
Bartlett's test		Chi square	3.47e+05	
		df	6	
		Sig.	0.000	

Note: Tourism expenditure (TE), Tourism arrivals (TA), and Tourism revenue (TR).

Table B2. Baseline regressions using GLS estimator (with weighted tourism index).

	GLOBAL	HICs	UMICs	LMICs	LICs
TAI	−0.0346*	−0.0257	−0.1154*	−0.1245***	0.7834***
	(0.0151)	(0.0197)	(0.0495)	(0.0343)	(0.1508)
LogGDP	−0.1290***	−0.1385***	−0.0458*	−0.1234***	−0.2692***
	(0.0050)	(0.0062)	(0.0214)	(0.0098)	(0.0370)
TAX	−0.0501**	−0.0188	−0.0177	−0.1205*	−0.8106***
	(0.0166)	(0.0195)	(0.0519)	(0.0501)	(0.0912)
RL	0.0018*	0.0017	0.0040	−0.0045*	−0.0210
	(0.0009)	(0.0010)	(0.0040)	(0.0019)	(0.0171)
Constant	1.4580***	1.6074***	0.8148***	1.4628***	1.8599***
	(0.0403)	(0.0635)	(0.1603)	(0.0798)	(0.2327)
Observations	1596	861	273	420	42
N	76	41	13	20	2
X^2	28522.03	11707.03	1979.68	13868.33	2616.82
Year fix effects	Yes	Yes	Yes	Yes	Yes
Country fix effects	Yes	Yes	Yes	Yes	Yes

The dependent variable is informal economy. The asterisks in the parameters mean: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. TAI is the tourism activity index build with PCA.

Table B3. Cross-sectional dependence test and slope homogeneity test (with weighted tourism index).

Panel A: CD test. Cross section dependency tests of Pesaran (2004)					
Variables	IEC	TAI	LogGDP	TAX	RL
Test statistic	21.44***	6.10***	6.41***	20.60***	7.54***
p-value	0.000	0.0000	0.0000	0.0000	0.0000
Panel B: CD test. Test of Pesaran (2015) for weak cross-sectional dependence					
Variables	IEC	TAI	logGDP	TAX	RL
Test statistic	55.12***	29.24***	41.44***	51.11***	50.15***
p-value	0.000	0.0000	0.0000	0.0000	0.0000
Panel C: Slope homogeneity tests	$\tilde{\Delta}$	$\tilde{\Delta}_{adj}$			
Test statistic	25.99***	30.31***			
p-value	0.000	0.000			

The asterisks in the parameters mean: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. TAI is the tourism activity index build with PCA.

Table B4. Results of the second-generation unit root test (with weighted tourism index).

		Variables				
Group	Test	IEC	TAI	logGDP	TAX	RL
Levels						
Global	CIPS	−0.649	−0.983	−0.532	−0.622	−0.071
	CADF	−0.805	−0.214	−0.549	−0.162	0.085
HICs	CIPS	−0.492	−0.877	−0.709	−0.865	−0.145
	CADF	−0.828	−0.228	−1.57	−0.849	1.547
HMICs	CIPS	−1.778	−0.822	−0.967	−1.56	−0.347
	CADF	−0.674	−1.332	−0.052	−0.221	−0.403
LMICs	CIPS	−0.76	−0.025	−0.731	−0.949	−0.334
	CADF	−0.848	−1.257	−0.788	−0.615	0.483
LICs	CIPS	−0.191	−0.742	−0.421	−0.46	1.747
	CADF	−0.133	−1.291	−1.084	−0.307	1.547
First differences						
Global	CIPS	−3.786***	−3.372***	−2.669***	−3.759***	−2.208**
	CADF	−1.942**	−2.603***	−2.686***	−3.299***	7.948***
HICs	CIPS	−3.629***	−3.266***	−2.846***	−4.002***	−4.282***
	CADF	−1.965**	−0.617***	−1.707**	−2.986***	3.410***
HMICs	CIPS	−3.915***	−3.211***	−3.104***	−3.697	−3.484***
	CADF	−1.811	−3.805***	−3.189***	−2.358***	−3.540***
LMICs	CIPS	−3.897***	−4.498***	−2.868***	−4.086***	−2.471**
	CADF	−1.985**	−2.73***	−1.925	−3.752***	4.346***
LICs	CIPS	−4.327***	−6.215*	−3.558***	−2.597**	2.610***
	CADF	−2.270**	−3.364***	−2.221**	−2.444**	2.810***

The asterisks in the parameters mean: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. TAI is the tourism activity index build with PCA.

Table B5. Results of the Westerlund (2007) co-integration test (with weighted tourism index).

		TAI		logGDP		TAX		RL	
Group	Estadístic	*Valor*	*p-value*	*Valor*	*p-value*	*Valor*	*p-value*	*Valor*	*p-value*
Global	Gt	−1.131	0.000	−3.247	0.000	−4.547	0.000	−4.557	0.000
	Ga	−11.122	0.000	−21.997	0.000	−1.997	0.000	−56.543	0.000
	Pt	−21.161	0.000	−42.875	0.000	−2.764	0.000	−44.175	0.000
	Pa	−12.564	0.000	−24.516	0.000	−55.32	0.000	−4.516	0.000
HICs	Gt	−2.311	0.000	−5.021	0.000	−5.021	0.000	−5.021	0.000
	Ga	−16.266	0.000	−37.795	0.000	−33.543	0.000	−3.795	0.000
	Pt	−10.126	0.000	−31.278	0.000	−11.323	0.000	−1.278	0.000
	Pa	−11.540	0.000	−34.629	0.000	−54.234	0.000	−3.122	0.000
UMICs	Gt	−2.546	0.000	−4.708	0.000	−3.528	0.000	−12.876	0.000
	Ga	−12.303	0.000	−35.280	0.000	−20.252	0.000	−14.322	0.000
	Pt	−16.664	0.000	−25.633	0.000	−23.455	0.000	−5.987	0.000
	Pa	−11.301	0.000	−35.021	0.000	−4.231	0.000	−5.436	0.000
LMICs	Gt	−6.116	0.000	−4.876	0.000	−8.423	0.000	−7.097	0.000
	Ga	−21.306	0.000	−37.048	0.000	−77.043	0.000	−6.867	0.000
	Pt	−14.101	0.000	−25.102	0.000	−24.434	0.000	−21.766	0.000
	Pa	−22.113	0.000	−41.608	0.000	−33.675	0.000	−12.808	0.000
LICs	Gt	−6.141	0.000	−5.202	0.000	−4.665	0.000	−17.302	0.000
	Ga	−20.456	0.000	−40.982	0.000	−1.243	0.000	−65.992	0.000
	Pt	−16.066	0.000	−15.335	0.000	−3.351	0.000	−89.385	0.000
	Pa	−16.101	0.000	−34.258	0.000	−12.234	0.000	−4.432	0.000

The *p*-value was obtained through the bootstrapping method with 200 repetitions. TAI is the tourism activity index build with PCA.

2016. His thesis examined the effects of new business formation on the productivity of manufacturing establishments. His research interest includes informal economic activities, entrepreneurship, and technology transfer.

Appendix A

Appendix B

Table B6. FMOLS and DOLS analysis (with weighted tourism index).

	FMOLS					DOLS				
	GLOBAL	HICs	UMICs	LMICs	LICs	GLOBAL	HICs	UMICs	LMICs	LICs
TAI	−0.55***	−0.27***	−0.21***	−0.27***	0.11***	−0.81***	−0.33***	−0.18***	−0.13***	0.08***
	(−18.45)	(−31.76)	(−13.53)	(−3.23)	(4.62)	(−13.21)	(−11.23)	(−14.33)	(−5.11)	(5.34)
logGDP	−0.06***	−0.04***	−0.02***	0.02***	0.17***	−0.034***	−0.032***	−0.018***	0.023***	0.19***
	(−22.22)	(−34.57)	(−164.57)	(57.31)	(4.12)	(−14.31)	(−22.45)	(−101.13)	(41.22)	(3.22)
TAX	−0.11***	−0.02***	−0.37***	0.22***	−0.19**	−0.15***	−0.041***	−0.26***	0.24***	−0.22*
	(−9.21)	(−8.79)	(−31.59)	(−5.13)	(−2.12)	(−4.15)	(−4.35)	(−33.12)	(−4.16)	(−1.98)
RL	−0.12***	−0.06***	−0.02***	−0.17***	−0.35***	−0.18***	−0.052***	−0.031***	−0.19***	−0.26***
	(−5.11)	(−121.38)	(−35.12)	(−46.87)	(9.11)	(−5.22)	(−132.43)	(−43.14)	(−38.34)	(4.33)
Observations	1596	861	273	420	42	1596	861	273	420	42
N	76	41	13	20	2	76	41	13	20	2
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

The dependent variable is informal economy. The asterisks in the parameters mean: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. t-statistic in parenthesis. TAI is the tourism activity index build with PCA.

Table B7. Results from Co-integration coefficients FMOLS long-run analysis for the HICs (with weighted tourism index).

	Dependent variable: Informal economy							
Country	TAI	*t*-stat	TAX	*t*-stat	LogGDP	*t*-stat	RL	*t*-stat
Australia	−0.24	−7.8	−0.1	−3.18	0.03	0.51	−0.01	10.48
Austria	0.19	4.33	−0.09	−1.99	−0.11	−12.3	0.07	11.34
Bahamas	−0.8	−29.9	0.49	7.83	0.16	24.59	0.24	36.14
Belgium	−0.27	−9.4	0.05	0.31	0.01	4.02	−0.02	2.88
Canada	−0.31	−25.82	−0.4	−9.51	0.03	17.94	−0.01	22.55
Chile	−0.29	−7.67	−0.43	−5.62	0.02	4.2	−0.03	0.85
Croatia	−0.45	−46.82	0.33	2.91	0.18	40.9	−0.12	−22.17
Cyprus	0.16	10.71	−0.17	−18.03	−0.06	−2.84	0.05	2.37
Czech Republic	−0.3	−37.72	0.26	2.24	0.02	23.71	−0.03	10.2
Denmark	0.16	18.59	−0.28	−9.16	−0.44	−26.84	0.66	26.03
Estonia	−0.23	−10.97	0.67	18.03	−0.19	−36.88	0.35	50.03
Finland	0.22	8.36	−0.29	−12.03	−0.17	−52.3	0.2	60.41
France	−0.28	−39.71	−0.51	−12.33	0.04	35.75	−0.05	−1.16
Germany	−0.17	−13.48	−1.56	−7.87	0.04	14.28	−0.04	4.79
Greece	−0.21	−37.58	0.99	37.35	0.01	47.4	−0.05	0.98
Hungary	−0.32	−20.71	0.5	9.67	0.04	21.46	−0.03	24.91
Iceland	−0.13	−15.04	−0.12	−21.4	−0.08	−20.88	0.05	31.43
Ireland	−0.22	−5.16	0.28	11.43	−0.09	−10.89	0.06	12.88
Israel	−0.25	−10.56	−0.03	−0.47	−0.17	−39.57	0.27	48.91
Italy	−0.21	−2.39	0.43	3.56	0.02	2.89	−0.05	−4.18
Japan	−0.13	−7.36	−0.39	−3.53	0.01	6.91	−0.06	−2.78
Korea. Rep.	−0.34	−123.08	0.01	0.13	0.09	34.29	−0.05	−0.37
Kuwait	0.19	2.99	4.34	3.25	−0.04	−1.63	−0.04	0.97
Latvia	0.17	10.57	−0.54	−22.28	−0.26	−31.37	0.38	42.66
Lithuania	0.2	1.11	−0.12	−4.15	−0.22	−27.74	0.42	41.29
Luxembourg	0.18	125.42	−0.05	−2.16	−0.14	−177.76	0.13	199.06
Malta	−0.25	−8.06	0.03	0.87	0.21	13.55	−0.04	4.57
Netherlands	−0.23	−2.79	0.06	0.53	−0.13	−6.38	0.16	7.4
New Zealand	−0.24	−27.09	0.23	26.62	0.01	16.76	−0.06	−6.84
Norway	−0.21	−0.78	−0.38	−7.72	−0.08	−2.4	0.06	1.22
Poland	−0.32	−9.59	2.79	31.24	0.05	10.32	−0.1	−7.58
Portugal	−0.21	−15.01	−0.5	−9.21	0.01	22.67	0.04	5.12
Singapore	−0.14	−26.17	−0.18	−6.24	0.02	24.13	0.03	15.27
Slovak Rep.	−0.27	−39.31	0.15	2.97	0.01	32.09	0.05	0.14
Slovenia	−0.47	−27.7	−2.33	−25.36	0.23	24.13	−0.17	−16.24
Spain	−0.26	−35.5	−0.23	−11.22	0.02	44.75	0.04	26.57
Sweden	0.25	7.25	−0.26	−10.85	−0.27	−20.1	0.34	24.05
Switzerland	−0.13	−27.98	−0.77	−36.69	0.02	49.91	0.05	2.56
Trin.and Tobago	−0.23	−2.61	−0.34	−5.81	0.01	4.85	0.02	29.19
UK	−0.20	−0.32	−0.32	−5.44	−0.04	−1.05	0.03	6.25
United States	−0.11	−13.11	−0.48	−25.11	0.02	16.37	−0.04	14.06
Uruguay	−0.42	−62.43	0.11	0.06	0.05	19.67	0.16	33.46
Panel results	−0.27	−31.76	0.02	8.79	−0.04	34.57	−0.06	−121.3

TAI is the tourism activity index build with PCA.

Table B8. Results from co-integration coefficients FMOLS long-run analysis for UMICs (with weighted tourism index).

	Dependent variable: Informal economy							
Country	TAI	*t*-stat	TAX	*t*-stat	LogGDP	*t*-stat	RL	*t*-stat
Argentina	−0.29	−22.31	−0.86	−11.05	0.02	45.8	−0.06	−3.31
Belarus	0.08	20.37	0.06	3.14	−0.36	−17.2	0.54	30.54
Brazil	−0.30	−37.41	0.01	4.27	−0.03	−3.58	0.06	5.11
Bulgaria	0.15	3.82	0.02	3.27	−0.05	−4.19	0.02	15.59
Costa Rica	−0.31	−37.26	−0.45	−4.15	0.02	78.38	−0.07	−2.37
Dominican Rep.	−0.43	−35.04	0.45	11.39	0.11	47.54	−0.07	−5.85
Guatemala	−0.11	−4.15	0.24	2.8	0.06	15.82	0.08	1.47
Iran. Islamic Rep.	−0.24	−0.46	0.13	7.61	0.05	7.65	0.04	9.9
Jamaica	−0.31	−0.51	0.01	3.25	0.06	17.28	−0.08	−1.27
Jordan	−0.29	−56.98	−0.45	−27.99	0.01	39.03	0.03	18.62
Kazakhstan	−0.31	−1.87	−0.93	−10.93	0.03	8.81	0.03	12.68
Malaysia	−0.22	−46.29	−0.37	−8.61	0.02	72.5	0.06	6.36
Mexico	−0.26	−4.69	0.33	7.77	−0.01	−23.99	0.05	19.32
Namibia	0.19	6.79	0.44	8.46	−0.11	−1.56	0.04	17.78
Peru	−0.14	−55.98	−1.58	−9.6	0.18	63.61	−0.1	−2.35
Romania	−0.33	−38.37	0.24	0.93	0.03	34.51	0.04	17.08
Russia	−0.46	−33.51	−1.95	−22.05	−0.2	−46.53	0.04	9.28
South Africa	−0.32	−41.99	−0.93	−17.06	−0.06	−102.17	−0.06	−2.28
Sri Lanka	−0.33	−1.77	0.39	4.02	−0.05	−10.32	0.06	3.2
Thailand	−0.31	−66.9	−0.98	−39.2	−0.09	−130.85	0.05	45.74
Panel results	−0.21	−13.53	−0.37	−31.59	−0.02	−164.57	−0.02	−35.1

TAI is the tourism activity index build with PCA.

Table B9. Results from Co-integration coefficients FMOLS long-run analysis for LMICs (with weighted tourism index).

	Dependent variable: Informal economy							
Country	TAI	*t*-stat	TAX	*t*-stat	LogGDP	*t*-stat	RL	*t*-stat
Angola	−0.12	−11.33	−0.16	−2.04	0.04	15.62	0.08	1.74
Egypt	−0.19	−12.13	−0.15	−2.16	0.52	9.19	−0.02	−1.27
El Salvador	−0.25	−22.41	−0.31	−1.49	0.33	10.12	−0.03	−3.40
India	−0.48	−4.27	−0.43	−1.36	0.44	5.25	0.04	1.80
Indonesia	−0.09	−54.21	−0.46	−3.50	0.11	8.69	−0.01	−6.20
Mongolia	−0.12	−5.23	0.09	4.81	−0.09	−30.85	0.14	34.01
Morocco	−0.34	−5.32	0.15	2.37	0.03	4.21	0.04	9.17
Myanmar	−0.21	−4.52	3.66	4.16	0.05	3.86	0.06	8.21
Nicaragua	−0.16	−4.15	0.22	3.24	0.04	15.03	−0.07	−7.20
Philippines	−0.15	−2.77	−0.62	−3.44	0.13	16.05	0.03	30.70
Tunisia	−0.20	−4.66	0.43	5.32	0.33	4.08	0.08	1.62
Ukraine	0.44	1.99	0.22	4.51	−0.30	−8.00	0.81	66.10
Zambia	0.43	2.43	−0.19	−5.41	−0.36	−24.42	0.94	29.27
Panel results	−0.27	−3.23	0.22	−5.13	0.02	57.31	−0.17	−46.87

TAI is the tourism activity index build with PCA.

Table B10. Results from Co-integration coefficients FMOLS long-run analysis for the LICs (with weighted tourism index).

	Dependent variable: Informal economy							
Country	TAI	t-stat	TAX	t-stat	LogGDP	*t*-stat	RL	t-stat
Ethiopia	0.32	4.21	−0.33	−3.11	0.09	4.16	0.18	9.11
Mali	0.12	3.11	0.06	0.55	0.14	5.11	−0.55	−5.32
Panel results	0.11	4.62	−0.19	−2.12	0.17	4.12	−0.35	−9.11

TAI is the tourism activity index build with PCA.

Table B11. The results of Dumitrescu and Hurlin (2012) panel causality test (with weighted tourism index).

Null hypothesis	Group	Z-bar	*p*-value	Result
TAI ≠ > IEC	Global	2.3355*	0.0315	Causality relationship
	HICs	11.331	0.1321	No causality relationship
	UMICs	1.5013*	0.0112	Causality relationship
	LMICs	−11.132*	0.0122	Causality relationship
	LICs	2.0302**	0.0010	Causality relationship
IEC ≠ > TAI	Global	2.3241***	0.0005	Causality relationship
	HICs	1.3251*	0.0331	Causality relationship
	UMICs	2.3122***	0.0001	Causality relationship
	LMICs	−0.1352	0.2233	No causality relationship
	LICs	11.555	0.1032	No causality relationship
logGDP ≠ > IEC	Global	8.1429***	0.0000	Causality relationship
	HICs	8.9692***	0.0000	Causality relationship
	UMICs	3.2538***	0.0011	Causality relationship
	LMICs	−0.0440	0.9649	No causality relationship
	LICs	0.4477	0.6544	No causality relationship
IEC ≠ > logGDP	Global	6.9230***	0.0000	Causality relationship
	HICs	8.1847***	0.0000	Causality relationship
	UMICs	0.8221	0.4110	No causality relationship
	LMICs	16.386	0.1013	No causality relationship
	LICs	−10.279	0.3040	No causality relationship
TAX ≠ > IEC	Global	7.4937***	0.0000	Causality relationship
	HICs	6.2288***	0.0000	Causality relationship
	UMICs	4.3819***	0.0000	Causality relationship
	LMICs	2.1587*	0.0309	Causality relationship
	LICs	−0.9129	0.3613	No causality relationship
IEC ≠ > TAX	Global	3.4133***	0.0006	Causality relationship
	HICs	4.2066***	0.0000	Causality relationship
	UMICs	0.0752	0.9401	No causality relationship
	LMICs	0.5638	0.5729	No causality relationship
	LICs	0.4083	0.6830	No causality relationship
RL ≠ > IEC	Global	0.001	0.98	No causality relationship
	HICs	0.001	0.98	No causality relationship
	UMICs	17.903	0.0734	No causality relationship
	LMICs	0.001	0.98	No causality relationship
	LICs	0.001	0.98	No causality relationship
IEC ≠ > RL	Global	0.001	0.98	No causality relationship
	HICs	0.001	0.98	No causality relationship
	UMICs	2.7338**	0.0063	Causality relationship
	LMICs	0.001	0.98	No causality relationship
	LICs	0.001	0.98	No causality relationship

The asterisks in the parameters mean: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. ≠ > means variable x does not Granger cause variable. TAI is the tourism activity index build with PCA.

Table B12. Result of the VECM model (with weighted tourism index).

Group	Dependent variable	Short-run relationship					Long-run relationship
		IEC	TAI	LogGDP	TAX	RL	ECT
GLOBAL	IEC	–	−0.002***(0.000)	−0.041***(0.000)			−0.172**(0.001)
	TAI	−1.09***(0.000)	–	1.017***(0.000)	0.441***(0.000)	−0.006 (0.19)	−0.191***(0.000)
	LogGDP	−1.10***(0.000)	0.054***(0.000)	–	0.120***(0.000)	0.002***(0.000)	−0922***(0.000)
	TAX	−0.09***(0.000)	0.015***(0.000)	0.071***(0.000)	–	−0.001***(0.000)	−0.061***(0.000)
	RL	−0.51 (0.400)	0.051***(0.000)	0.074***(0.000)	0.051***(0.000)	–	−0.002* (0.023)
HICs	IEC	–	−0.004 (0.128)	−0.018***(0.000)	−0.25** (0.005)	0.022 (0.320)	−0.214***(0.000)
	TAI	−1.31** (0.040)	–	0.014***(0.000)	0.444***(0.000)	−0.013**(0.029)	−0.191* (0.041)
	LogGDP	−1.87***(0.000)	0.044 (0.052)	–	0.220 (0.200)	0.003***(0.000)	0641***(0.000)
	TAX	−0.09***(0.000)	0.021***(0.000)	0.011***(0.000)	–	−0.002 (0.210)	−0.044***(0.000)
	RL	−0.07***(0.000)	0.058***(0.000)	0.044***(0.000)	0.068 (0.000)	–	−0.014* (0.041)
UMICs	IEC	–	−0.004***(0.000)	−0.014*(0.041)	−0.144***(0.000)	0.021* (0.040)	−0.414***(0.000)
	TAI	−1.17**(0.010)	–	1.014***(0.000)	0.225 (0.414)	−0.012 (0.16)	−0.161***(0.000)
	LogGDP	−1.99 (0.900)	0.042***(0.000)	–	0.120 (0.000)	0.002 (0.740)	0.611***(0.000)
	TAX	−0.044*(0.041)	0.015 (0.241)	0.044 (1.110)	–	−0.001***(0.000)	−0.026 (0.026)
	RL	−0.09***(0.000)	0.015***(0.000)	0.047***(0.000)	0.064 (0.000)	–	0.114***(0.000)
LMICs	IEC	–	−0.002***(0.001)	−0.011 (0.508)	−0.152***(0.000)	0.003 (0.300)	−0.114***(0.000)
	TAI	−1.55 (0.110)	–	1.117***(0.000)	0.544 (1.200)	−0.014 (0.12)	−0.121***(0.000)
	LogGDP	−1.540 (0.090)	0.044***(0.000)	–	0.120* (0.020)	0.006***(0.000)	−0948 (0.220)
	TAX	−0.88 (0.900)	0.025 2 (0.055)	0.044 (0.110)	–	−0.001 (0.900)	−0.069** (0.010)
	RL	−0.51 (0.400)	0.041***(0.000)	0.041***(0.000)	0.011***(0.000)	–	0.444***(0.000)
LICs	IEC	–	−0.004*(0.041)	−0.011 (0.077)	−0.155 (0.065)	0.002 (0.320)	−0.444***(0.000)
	TAI	1.09 (0.057)	–	1.441 (0.140)	0.442 (0.520)	−0.006***(0.000)	0.226* (0.018)
	LogGDP	−0.77 (0.998)	0.081***(0.000)	–	0.120* (0.024)	0.007 (0.210)	−0114***(0.000)
	TAX	−0.04 (0.080)	0.015* (0.021)	0.001***(0.000)	–	−0.021***(0.000)	0.224* (0.041)
	RL	−0.81 (0.740)	0.015 (0.410)	0.071***(0.000)	0.042 (0.000)	–	−0.064 (0.066)

The asterisks in the parameters mean: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. TAI is the tourism activity index build with PCA.

Table B13. The results of Dumitrescu and Hurlin (2012) panel causality test for independent variables (with weighted tourism index).

Null hypothesis	Group	Z-bar	*p*-value	Result
TAI ≠>logGDP	Global	1.9311*	0.023	Causality relationship
	HICs	1.445	0.154	No causality relationship
	UMICs	1.1014*	0.015	Causality relationship
	LMICs	−11.144*	0.014	Causality relationship
	LICs	4.0404**	0.001	Causality relationship
logGDP ≠>TAI	Global	4.12344***	0.000	Causality relationship
	HICs	1.4412*	0.045	Causality relationship
	UMICs	4.4144***	0.000	Causality relationship
	LMICs	−0.1414	0.213	Causality relationship
	LICs	11.3123***	0.004	No causality relationship
TAI ≠>TAX	Global	4.1311***	0.000	Causality relationship
	HICs	4.4241***	0.000	Causality relationship
	UMICs	4.6724***	0.001	Causality relationship
	LMICs	−0.0440	0.424	No causality relationship
	LICs	0.4455	0.214	No causality relationship
TAX ≠>TAI	Global	1.4140***	0.000	Causality relationship
	HICs	4.1445***	0.000	Causality relationship
	UMICs	0.4111	0.410	No causality relationship
	LMICs	2.442	0.104	No causality relationship
	LICs	−10.154	0.440	No causality relationship
TAI ≠>RL	Global	5.42345***	0.000	Causality relationship
	HICs	2.1144***	0.000	Causality relationship
	UMICs	4.41614***	0.000	Causality relationship
	LMICs	1.1145*	0.040	Causality relationship
	LICs	−0.4114	0.221	No causality relationship
RL ≠>TAI	Global	4.4144***	0.000	Causality relationship
	HICs	4.1022***	0.000	Causality relationship
	UMICs	0.0511	0.440	No causality relationship
	LMICs	0.1244	0.151	No causality relationship
	LICs	0.4044	0.244	No causality relationship
logGDP≠>TAX	Global	0.002*	0.048	Causality relationship
	HICs	0.002	0.480	No causality relationship
	UMICs	27.403	0.073	No causality relationship
	LMICs	0.002	0.486	No causality relationship
	LICs	0.042***	0.000	Causality relationship
TAX ≠>logGDP	Global	0.012	0.331	No causality relationship
	HICs	0.031	0.482	No causality relationship
	UMICs	2.7338**	0.006	Causality relationship
	LMICs	0.002	0.223	No causality relationship
	LICs	0.002	0.483	No causality relationship
logGDP≠> RL	Global	7.4437***	0.000	Causality relationship
	HICs	6.2288***	0.000	Causality relationship
	UMICs	4.3824***	0.000	Causality relationship
	LMICs	2.2187*	0.030	Causality relationship
	LICs	−0.4224	0.362	No causality relationship

(continued)

Table B13. (continued)

Null hypothesis	Group	Z-bar	*p*-value	Result
RL ≠>logGDP	Global	3.4233***	0.000	Causality relationship
	HICs	4.2066***	0.000	Causality relationship
	UMICs	0.0712	0.440	No causality relationship
	LMICs	0.1638**	0.024	Causality relationship
	LICs	0.4083	0.683	No causality relationship
RL ≠>TAX	Global	0.002**	0.023	Causality relationship
	HICs	0.002	0.484	No causality relationship
	UMICs	7.403**	0.003	Causality relationship
	LMICs	1.002	0.481	No causality relationship
	LICs	0.002***	0.001	Causality relationship
TAX≠> RL	Global	0.002	0.482	No causality relationship
	HICs	0.002	0.323	No causality relationship
	UMICs	2.7338**	0.006	Causality relationship
	LMICs	0.002	0.481	No causality relationship
	LICs	0.002	0.338	No causality relationship

The asterisks in the parameters mean: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. ≠ > means variable x does not Granger cause variable. TAI is the tourism activity index build with PCA.